Children First, Housing Later*

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October 10, 2026

Abstract

Should housing assistance require sobriety? A general-equilibrium model of homelessness, drug use, drug prices and mortality is calibrated to US data. Use concentrates among low-income households because a substance use disorder's mortality risk, unlike its unemployment risk, is uninsurable. Drug use harms users' children, cutting their consumption and, through a parent's disorder, their adult earnings. Sobriety conditions reduce these harms more than they expose screened-out users' children to homelessness. Due purely to the externalities on children, conditionality raises a newborn's welfare by 5.04 percent of lifetime consumption, mostly in childhood. Along the transition, childhood homelessness rises by up to 13 percent, but children born at the reform still gain 2.7 percent. Moral hazard alone is not the reason for conditionality: when isolated, conditionality lowers newborns' welfare by 1.31 percent, even when savings fund more vouchers.

Keywords: homelessness; housing vouchers; substance use disorder; intergenerational spillovers; externalities.

JEL classification: I18; I38; H23; R21; J62.

1 Introduction

At least 771,480 people were sleeping in shelters or on the street in the point-in-time (PiT) count of January 2024 performed by the Department of Housing and Urban Development (HUD), the most since the count began. In July 2025 an executive order directed the federal agencies to end support for Housing First policies that, in its words, "fail to promote treatment, recovery, and self-sufficiency".[1] Housing First gives low-income households housing assistance with no requirement to stop using drugs; the alternative, Treatment First, makes the assistance conditional on sobriety. The funding notice that HUD issued after the order would have capped permanent housing, through which most Housing First assistance is given, at 30 percent of renewal funding, down from more than 80 percent, and would have opened the rest to transitional housing that can require treatment, had a federal court not struck the notice down.[2]

The debate is usually framed as a contest between two behavioural claims about the households who receive the assistance.[3] Critics of Housing First say that unconditional assistance creates moral hazard: it weakens the incentive to stay sober and employed, so drug use and substance use disorders (SUD) rise, and

*I thank Ayşe İmrohoroğlu, Kai Zhao, Joseph Steinberg, Efraim Benmelech, Jonathan Alschech, Ryan Quandt, Christopher Sharp and the seminar participants at the University of Toronto for their helpful comments. The data and replication code are available on GitHub at `https://github.com/srotberg/children_first_housing_later`.

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[1] Executive Order 14321, *Ending Crime and Disorder on America's Streets*, 24 July 2025, *Federal Register* 90, 35817–35820.

[2] HUD, *FY 2025 Continuum of Care Competition and Youth Homeless Demonstration Program Grants*, FR-6900-N-25, 13 November 2025; on the shares of funding, Congressional Research Service, Insight IN12626. The ruling is *State of Washington v. U.S. Department of Housing and Urban Development*, No. 1:25-cv-00626 (District of Rhode Island, 29 June 2026), which set aside both the November and the December 2025 notices. See `https://www.npr.org/2025/11/14/nx-s1-5553561/homelessness-housing-funding-trump-administration-hud`; on the court's ruling, `https://marylandmatters.org/2026/06/30/ri-federal-judge-strikes-down-huds-2025-restrictions-for-housing-first-initiatives/`.

[3] For the case against Housing First as the federal government now makes it, see the executive order cited above, and on its incentives and costs, Eide (2020); for the case for it, see the National Alliance to End Homelessness, `https://endhomelessness.org/blog/ceo-corner-week-of-july-28/`, and on housing stability and costs, Tsemberis et al. (2004) and Larimer et al. (2009).

with them homelessness, creating a vicious circle, with the cost falling on others who fund the programme. Critics of Treatment First say that conditionality delays stable housing for those who need it most, because recovering while homeless is harder: here, the argument is that the circle runs the other way, from homelessness to more SUD and back, so in fact, Housing First improves the budget. Randomised trials of Housing First speak to both claims: they find large gains in housing stability and mixed-to-null effects on drug use by following the participants for at most three years (Section 2). However, they do not follow them decades into the reform, or follow the impacts on government budgets and the children of the participants. A model, in contrast, can be used to estimate those.

I weigh in on the debate by building a quantitative life-cycle model in which households decide whether to use drugs knowing the risk of acquiring a SUD, and in which drug use harms not only the user but also the user's children. I begin by showing that the moral hazard that dominates the debate is real but does not drive the answer at all. To isolate it, I remove the externalities that harm children, leaving only the change in behaviour and its impact on the budget. A newborn is then worse off in the long run under Treatment First, by 1.31 percent in consumption-equivalent terms, even when the screen's savings on vouchers and the higher revenues from more employment are spent on more vouchers provided to other, non-user households. This is because, without externalities, conditionality removes an option from households who weigh the risks of drug use correctly, as the evidence suggests they do (Section 2), and gives them back significantly less through the government's budget than it takes.

I show that the answer turns instead on the children, through three externalities that a parent takes into account only through imperfect altruism towards the child: two that Housing First leaves in place and one that Treatment First creates. The two it leaves in place fall on the user's children. Children share their parents' consumption and dwelling but not the drug, so every unit a parent spends on drugs, and every unit of earnings lost while a SUD keeps them unemployed, is consumption and housing the child goes without: the *budget externality*. Furthermore, a child whose parent had a SUD while it lived at home starts adult life with lower productivity: the *earnings externality*. The one Treatment First creates runs through homelessness: by denying assistance to drug users, it exposes more of them to the shelter, and a child lives wherever its parent lives: the *childhood homelessness externality*.

Whose children bear these costs depends on who uses drugs. Use is concentrated at the bottom of the income distribution, where the average user's income is about 74 percent of the average, and most households do not use at all. The literature has explained the two facts together only by assuming that lower-education households have a systematic preference for drugs (Greenwood et al., 2025). I model drugs as a luxury good bought at a price, which puts most households at a corner where they consume none. On its own, that would concentrate use at the *top*; what reverses the gradient is the higher risk of dying that comes with a SUD, a loss that rises with income and that, unlike losing one's job, cannot be saved against, so in equilibrium use concentrates at the bottom. The model therefore explains both facts without assuming systematic differences in preferences.

I proceed with two models of rational forward-looking households. The first is a stylized model with two periods, in which a SUD raises the risks of dying and of losing one's job. Its sole purpose is to show why the income gradient in drug use reverses. The second is a quantitative general-equilibrium life-cycle model calibrated to the United States, whose purpose is to test whether Treatment First raises or lowers welfare relative to Housing First, and whether moral hazard or the externalities on children drive the answer. Its key ingredients are housing, with a shelter at the bottom of the housing ladder; rationed, means-tested vouchers; income and unemployment risk, with unemployment insurance and food assistance; retirement and social security; recovery, relapse and treatment; serious mental illness (SMI); children who share their parent's consumption and dwelling and start adult life with lower productivity if the parent had a SUD; a parent's imperfect altruism towards the child; a drug test that misses some users; a drug price that clears the drug market; and a government budget that balances through the probability of receiving a voucher.

The quantitative model lets me isolate the three externalities. I run the same reform with and without the externalities, with only the screen in effect, so the one difference between the two runs is whether drug use harms children. Without externalities a newborn loses 1.50 percent; with them it gains 5.04 percent, childhood included, while the share of children in a shelter rises only from 1.03 to 1.04 percent: the *budget and earnings externalities* dominate the *childhood homelessness externality*. Spending the screen's savings on more vouchers raises the gain to 13.11 percent, and letting the drug price clear as well, as it would after

a real reform, brings it down to 4.33 percent, since the lower drug price draws some households back into use. In the latter, the full general-equilibrium exercise, Treatment First cuts drug use by 32 percent and the share of households with a SUD by 33 percent in the long run, and 57 percent of the 4.33 percent gain comes in childhood.

On top of these long-run results, along the transition, homelessness and childhood homelessness first rise, by up to 15 and 13 percent, respectively, yet children born at the reform still gain 2.7 percent. Lastly, the screen's deterrence in the model is aligned with the best outside evidence I could find: of the users without a SUD that it tests, 59 percent stop when it is introduced, close to the 61 percent fall in marijuana use over four years among service members who stayed in the US military after it began testing at random and dismissing those who failed (Mehay and Pacula, 1999; Section 5.2).

2 Related literature

Randomised trials of Housing First. A large empirical literature evaluates programmes that provide rapid housing without conditioning on sobriety, and finds large and persistent improvements in housing stability and reductions in shelter and other public-service use (Tsemberis et al., 2004; Padgett et al., 2006; Larimer et al., 2009; Goering et al., 2014; Aubry et al., 2016). Evidence on substance use is mixed: most studies find no statistically significant change, and some report modest reductions in alcohol-related outcomes. Costantini (2025) finds that enrolling in Housing First lowers the mortality of homeless veterans with mental illness, while its Treatment First programmes have no lasting effect on it; Section 5.2 compares the model with it. These studies, however, measure outcomes over six to thirty-six months, which is short against a recovery process that Vaillant (1995) documents as gradual and protracted. A further limitation of a trial, and perhaps a deeper one, is its scope. A trial randomises households within a fixed environment, and holds fixed exactly the things that decide the welfare comparison: the price of the drug, the government's budget, and, most importantly, the effects on children, who are not in the sample.

Deterrence effects of zero-tolerance policies. The closest evidence on how much a drug test with a severe penalty deters drug use comes from the US military, which in 1981 began testing its personnel at random, 10 to 20 percent of a unit every month, and dismissing those who failed. Mehay and Pacula (1999) utilize this policy change to compare service members with civilians, following the same people over time: among those who stayed in the military, past-year marijuana use fell from 48.5 to 13.2 percent between 1980 and 1984, against a fall from 45.8 to 32.0 percent among those who stayed civilians, so the testing cut use among those subject to it by about 61 percent of what it would otherwise have been. Section 5.2 shows how the model performs on this dimension.

Economic models of drug use. Modelling substance use as forward-looking optimisation dates to Becker and Murphy (1988), and the assumption is empirically defensible at the margin where policy operates. Use responds to its price, even for drugs as habit-forming as heroin and cocaine, and more in the long run than in the short run, the signature of forward-looking choice rather than myopia (van Ours, 1995; Grossman and Chaloupka, 1998; Saffer and Chaloupka, 1999; Caulkins, 2001; Gallet, 2014). Greenwood et al. (2025), who allow perceived and true SUD risk to differ, find that the model-implied beliefs are actually very close to the true risk. Moreover, young people, if anything, overestimate the risk of becoming dependent, and those who perceive a higher risk drink less (Lundborg and Lindgren, 2002). A separate literature emphasises temptation and dynamic inconsistency (Laibson, 1997; Gruber and Kőszegi, 2001), under which present-biased households over-consume relative to their own long-run preferences. With such assumptions, Treatment First has a rationale even with no externalities at all. Therefore, that literature is complementary and cuts in the same direction: a present-biased household could value a sobriety requirement as a commitment device, so introducing it would strengthen the case for conditionality rather than weaken it. The paper also departs from Greenwood et al. (2025) in how it generates the income gradient in drug use. There, use concentrates among the less educated because their preferences for drugs differ; here no preference parameter varies with income, and the gradient comes from the uninsurable mortality risk of a SUD (Section 3).

General-equilibrium models of homelessness. I build directly on İmrohoroğlu and Zhao (2022), who model homelessness as a rational response to income risk, mental-health shocks, housing costs and housing policy (see also Abramson, 2026, on how eviction policy changes who becomes homeless, and Rotberg and Zhang, 2024, on rent control's effect on homelessness). In that framework housing policy reaches homelessness through income loss and housing costs alone, so a sobriety-contingent policy has no behavioural margin to act on in their model. Adding endogenous use, disorder and recovery, together with state-dependent mortality, creates that margin: the same voucher that insures a household also changes what using costs it, and the disorder it may acquire then feeds back into employment, housing and survival.[4]

Intergenerational transmissions. The *earnings externality* rests on what a parent's SUD does to a child. Parental drug abuse is recorded at more than a third of removals into foster care (AFCARS, FY2017), household substance abuse is one of the adverse childhood experiences of Felitti et al. (1998), and children of a parent with a SUD are more likely to acquire one themselves (Mellentin et al., 2016; Yoon et al., 2013). A parent who stays unemployed longer and spends on the drug has less to invest in the child, and that investment shows up in adult earnings: family income raises children's achievement (Dahl and Lochner, 2012), parental investment in education accounts for roughly half of the intergenerational correlation in earnings (Restuccia and Urrutia, 2004), and a year of schooling returns around ten percent (Card, 1999). Nilsson (2017) measures the loss from a parental substance shock directly, and the model's productivity penalty is set to it (Section 5). What this paper adds is not another estimate but a price: these effects sit inside an equilibrium model in which the parent's use decision is endogenous, so the transmitted loss is a quantity that policy moves rather than a coefficient.

Homelessness persists across generations too (Bassuk et al., 1997). In the model, this persistence comes from the persistence of earnings, whose intergenerational elasticity in the United States runs from about four tenths (Solon, 1992) to about six tenths (Mazumder, 2005), with the same persistence in ranks across the distribution (Chetty et al., 2014), and not from homelessness itself. A poor environment does lower a child's adult earnings beyond that (Chetty et al., 2016; Chyn, 2018), but the model imposes no penalty for growing up in a shelter. Adding one would matter little: under Treatment First the share of children in shelters does not rise in the long run, and during the transition the rise in childhood homelessness falls on at most 0.13 percent of children at any time, relative to eliminating parents' drug use for 5.01 percent of children in the first year, far too few to reverse the 2.7 percent welfare gain of the children born at the reform (Section 6.3).

Parental altruism towards children. How much of these costs a parent bears depends on the weight it puts on its child. Altonji et al. (1997) reject perfect altruism: when a dollar moves from a grown child to its parent, the parent gives back at most thirteen cents of it. The model has no grown children with budgets of their own, so that test cannot be run in it. Instead, the weight is set by behaviour. Unmarried adults living with one other adult and no child are 17.5 percent more likely to use drugs than lone parents with one child, of the same age, sex and family income, in the 2024 National Survey on Drug Use and Health (NSDUH), and the weight is the one at which removing a parent's concern for its child raises drug use in the model by as much (Section 5). The externality is then what the parent leaves out, one minus that weight, of what its choices cost the child.

That a drug user has children it affects is supported by the data. Drug users are nearly as likely as others to be parents: in the National Survey of Family Growth 2017–2019, 75 percent of adults aged 40 to 49 who used hard drugs in the past year had a child, against 81 percent of those who used none. They are only somewhat less likely to live with it.[5] Among adults aged 26 to 49, 40 percent of those with a drug use disorder have a child under 18 at home, against 48 percent of comparable adults who used no illicit drug. That gap is too small to matter, since even if all of the gain from Treatment First that comes in childhood shrank in proportion, the gain would be only 10 percent smaller. The model does overstate childhood homelessness, since every homeless parent takes its child into the shelter, while most homeless

[4]The paper does not focus on the disamenity that homelessness may impose on the rest of the population. In a contingent valuation, Loubière et al. (2020) find that Europeans would pay about 0.2 percent of average earnings a year to end homelessness. I also solved the model with a disamenity of that kind, and it barely changes the results.

[5]Users who have a child are less likely to live with it, partly because the state removes children: parental drug abuse is recorded at more than a third of removals into foster care, which is also a very poor environment for a child (AFCARS, FY2017).

adults have no child with them: shelters for single adults do not take children, so a homeless father's child usually stays with its mother or other relatives, and some children are taken into foster care. This, however, works against Treatment First.

The hazards and the drug market. Each of the model's hazards is anchored on, or checked against, a clinical or epidemiological estimate: the excess mortality of a SUD, of homelessness and of SMI (Plana-Ripoll et al., 2019; Heiberg et al., 2018; Meyer et al., 2023; Olfson et al., 2015), the onset of SMI and relapse under dual diagnosis (Nielsen et al., 2017; Bradizza et al., 2006), recovery (Vaillant, 1995; Milby et al., 2005), and the employment cost of losing housing (Desmond and Gershenson, 2016); Section 5 takes them one at a time. Making the drug price endogenous lets a policy that changes the number of users change what using costs everybody else. The model's demand elasticities are not calibrated and are compared with the estimates of Saffer and Chaloupka (1999), Grossman and Chaloupka (1998) and Olmstead et al. (2015) in Section 5.2; its supply elasticity is set from Mace et al. (2020) in Section 5.

3 A stylized model

In the data, fewer than one in ten people use drugs, and their use is concentrated among people with low incomes: the average user's income is 0.74 of the population average in the 2024 NSDUH. This section shows how both facts can be explained inside a simple economic model of drug use. There are two key ideas. Drugs are a luxury good, so most households consume none of them. And a SUD carries a risk of dying that cannot be insured against, a loss that grows with income, which deters high-income households and not low-income ones and so reverses the income gradient in drug use.

3.1 The model

There are two periods, adulthood and old age, and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty-six to eighty-five. Parameters with a natural annual counterpart are compounded to that length: the discount factor is β per period and $\bar{\phi}$ is survival over thirty years.

A household is characterised by its productivity y, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma_y^2, \sigma_y^2)$, normalised so that its mean is one. Everyone works in the first period. In the second, a household is out of work with probability u_0, or $u_1 > u_0$ if it has a SUD. Writing $e = 1$ for a household that works and $e = 0$ for one that does not, its earnings are $y\,e$: households out of work earn nothing, and there is no unemployment insurance. What it can do is save, at zero interest, in the first period; it cannot borrow.

The one transfer is food assistance, paid on each period's earnings as

$$F(y\,e) \;=\; \max\{0,\; \bar{c} - \varrho\, y\, e\}, \tag{1}$$

where $\bar{c}$ is the maximum benefit and ϱ the share of earnings by which it is reduced. Food assistance is in the model for one reason. To show the income reversal, Section 3.3 needs to solve the model with saving switched off, and a household out of work would have nothing to consume without food assistance, which with $\sigma > 1$ leaves its problem undefined; food assistance bounds its consumption from below.

Consumption c and drug consumption v form the bundle

$$X(c, v) = c^{1-\theta}(b + v)^{\theta}, \tag{2}$$

and flow utility is

$$u(c, v) \;=\; U\big(X(c, v)\big) \;+\; \aleph, \qquad U(X) = \frac{X^{1-\sigma}}{1-\sigma}. \tag{3}$$

The weight on the drug θ is its share of the whole, and σ is the coefficient of relative risk aversion.[6] Drugs can be bought only in the first period, at the price p_v. The pair (θ, b) governs the drug term: the marginal utility

[6] All the qualitative results of this section hold with log utility as well, with the model recalibrated to the same targets.

of the vth unit is proportional to $\theta/(b+v)$, so θ mostly sets how much a user uses and b whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, which is what generates non-users at all, because without the shifter every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish even when consumption is low.

Using drugs carries the risk of a SUD ($d = 1$ if a household has one and $d = 0$ if not), the same whatever the quantity,

$$\pi(v) \;=\; \pi_d \, \mathbf{1}\{v > 0\}. \tag{4}$$

The whole risk therefore sits on the decision to use at all, and θ alone sets how much a user uses once it has decided to actually use. Every household enters the first period without a SUD. The SUD is realised at the end of it, and the household survives into the second with probability

$$\phi(0) \;=\; \bar{\phi}, \qquad \phi(1) \;=\; \bar{\phi}^{\,\kappa_{\phi d}}, \tag{5}$$

where a SUD raises the probability of surviving the period to the power of $\kappa_{\phi d} > 1$, thus lowering it. The second period is the last and has no drug-use decision, so to a household with savings a and SUD status d it is worth

$$W(a, y, d) \;=\; (1 - u_d)\, U\big(X(y + a + F(y), 0)\big) \;+\; u_d\, U\big(X(a + \bar{c}, 0)\big) \;+\; \aleph. \tag{6}$$

In the first period, a household that does not use chooses its saving to maximize its lifetime utility,

$$V^0(y) \;=\; \max_{\{a \geq 0\}} \; \Big\{ U\big(X(y - a + F(y), 0)\big) + \aleph + \beta\, \bar{\phi}\, W(a, y, 0) \Big\}, \tag{7}$$

and one that uses chooses its saving and its drug consumption to maximize it,

$$V^1(y) \;=\; \max_{\{a \geq 0,\; v > 0\}} \; \Big\{ U\big(X(y - a + F(y) - p_v v,\; v)\big) + \aleph + \beta \Big[(1 - \pi_d)\, \bar{\phi}\, W(a, y, 0) + \pi_d\, \bar{\phi}^{\,\kappa_{\phi d}}\, W(a, y, 1) \Big] \Big\}. \tag{8}$$

It uses if $V^1(y) > V^0(y)$. A SUD reaches the second period through two channels: the mortality channel, which lowers the chance of living it, and the unemployment channel, which raises the chance of living it unemployed.[7]

3.2 Calibration

Table 1 lists what is set outside the model and Table 2 what is fitted inside it, where the five parameters are fitted jointly and each moment is matched exactly. Every target is documented in full in the replication package's data appendix, `data/DATA_APPENDIX.md`, which names the dataset behind each one, its year and universe, and the computation that turns it into the number quoted here. What follows gives the moment and its source; the appendix gives the arithmetic.

3.2.1 External calibration

Preferences and demographics are conventional: the discount factor over a thirty-year period is $\beta = 0.86$, which is 0.995 a year compounded,[8] risk aversion is $\sigma = 2$, as in the quantitative model, and survival over a period, the cumulative chance of surviving from age twenty-six to age fifty-five, is $\bar{\phi} = 0.93$, from the life table for the total population of the United States in 2010 (Arias, 2014). The chance that use causes a SUD, $\pi_d = 0.41$, is set so that the ratio of the SUD rate to participation matches the data in the 2024 NSDUH, 3.68 against 8.90 percent of adults aged 26 and over.

[7] The results in Section 3.3 hold also in a version of this model with housing, a shelter, unemployment insurance and housing vouchers. There, if a SUD has no consequences, the average user earns 3.54 times the population average; the risk of dying alone takes this to 0.52 at twice its calibrated strength, while the risk of losing one's job alone leaves it at 3.54. With saving switched off in that version, the risk of losing one's job alone takes it to 0.56.

[8] The annual discount factor here is higher than the one in the quantitative model, and this is deliberate. A period here is thirty years, so every consequence of using drugs arrives thirty years after the decision, whereas in the quantitative model consequences arrive annually and compound. A low annual discount factor would weigh that distant death much less and the mortality channel would be losing to impatience rather than to anything the model is about, so a higher factor corrects for the coarseness of the horizon.

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
β	Discount factor over a thirty-year period	0.995 annually, compounded	0.86
σ	Risk aversion	Standard; Attanasio and Weber (2010)	2
$\bar{\phi}$	Survival over a period, no SUD	US life table 2010, Arias (2014), ages 26–55	0.93
π_d	Chance that use starts a SUD	NSDUH 2024: SUD rate over participation	0.41
σ_y	Dispersion of log productivity	SCF 2022: bottom half holds 17.2% of earnings	0.95
u_0	Unemployment rate in period 2, no SUD	BLS CPS, average 2014–2024	0.049
u_1	Unemployment rate in period 2 with a SUD	NSDUH 2024, ages 26 and over	0.16
ϱ	Food assistance taper	SNAP benefit reduction rate	0.30
p_v	Price of a unit of drugs	Normalisation	1.00

Table 2: Internally calibrated parameters

Parameter	Target moment	Data	Model	Value
$\aleph$	VSL over income	80	80.0	8.64
b	Drug participation, NSDUH 2024	8.9%	8.9%	0.032
θ	Drug spending over users' income	16.0%	16.0%	0.22
$\kappa_{\phi d}$	Users' income over the average, NSDUH 2024	0.74	0.74	3.56
$\bar{c}$	Food assistance over labour income	0.42%	0.42%	0.10

The dispersion of log productivity σ_y is set so that the poorer half of the lognormal holds 17.2 percent of total income, as households aged 26 to 65 with positive wage and salary income do with their earnings in the 2022 Survey of Consumer Finances (SCF). In the second period, a household without a SUD is out of work with probability u_0, 4.9 percent, the average unemployment rate over 2014 to 2024, and one with a SUD with probability u_1, 16.25 percent, the unemployment rate of adults aged 26 and over with a SUD in the 2024 NSDUH. Food assistance tapers at $\varrho = 0.30$, the benefit reduction rate of the Supplemental Nutrition Assistance Program (SNAP), and the price of a unit of drugs p_v is normalised to one.

3.2.2 Internal calibration

The flow value of being alive $\aleph$ is matched to a value of a statistical life (VSL) of 80 times average annual income (Viscusi and Aldy, 2003).[9] The parameter b, which sets the marginal utility of the first unit of drugs, is chosen to match a drug participation rate of 8.90 percent from the 2024 NSDUH for respondents aged 26 and over with a disorder of any severity, and the weight on drugs θ is set to match the aggregate drug spending over the aggregate income of users, 16.0 percent, from retail spending in Midgette et al. (2019) and the incomes of users in the 2024 NSDUH. The level of food assistance $\bar{c}$ is set so that total food assistance is 0.42 percent of total labour income, the ratio of SNAP outlays to gross domestic product (GDP).

The mortality rate ratio $\kappa_{\phi d}$ is set to match the income of users, 0.74 of the population average in the 2024 NSDUH, because the death risk is what decides how far down the income distribution use sits, as Section 3.3 shows.

3.3 Who uses drugs

Drugs in the model are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. If a SUD had no consequences, that would be the only force operating, and drug use would concentrate at the top: with b refitted so that participation stays at 8.90 percent, the average user then earns 3.51 times the population average. What follows shows that it is the uninsurable risk of dying that reverses this gradient.

Figure 1 scales each of the two channels through which a SUD reaches old age, mortality and unemployment, from nothing to 200 percent of its calibrated value with the other switched off, by scaling $\kappa_{\phi d} - 1$ for the mortality channel and $u_1 - u_0$ for the unemployment channel. At every point b is refitted so that

[9] No household has a negative value of life in period 1, so death is never an optimal choice.

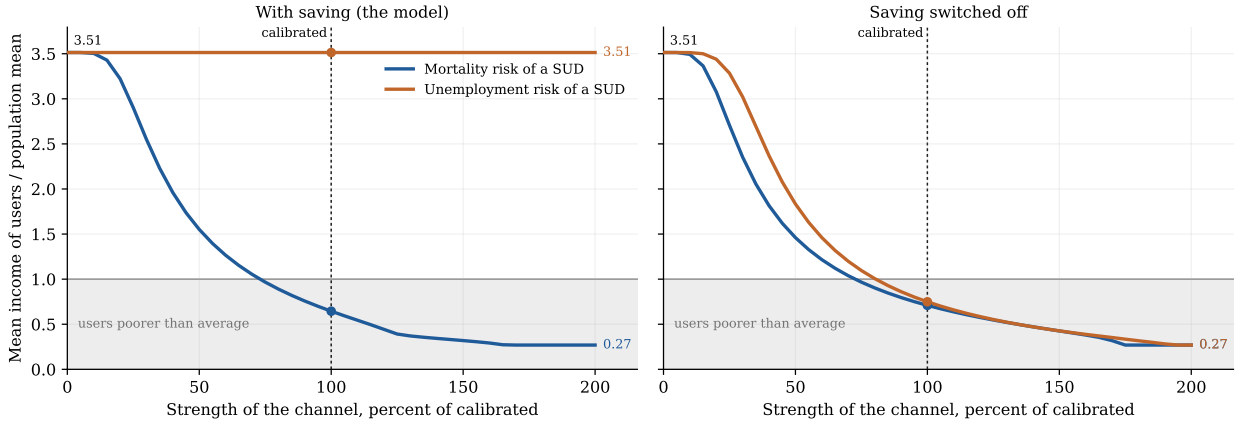


Figure 1: What it takes to reverse the income gradient in drug use

Notes: each curve scales one channel with the other switched off. Left: the model. Right: the same parameters with saving switched off. The vertical axis is average income of users over the population average, and b is recalibrated at every point so that participation is the calibrated 8.90 percent. The vertical dotted line marks the calibrated strength.

participation stays at its calibrated value: the amount of drug use is the same everywhere on the figure, and only who uses changes.

The left panel shows the model, which allows households to save. The mortality channel alone takes the average user from 3.51 times the population average down to 0.64 at its calibrated strength, crossing the average at 73 percent of that strength, and to 0.27 at 200 percent.[10] In contrast, the unemployment channel alone leaves the average user at 3.51 times the population average even at 200 percent: on its own it does not change who uses at all.

To understand why the risk of dying reverses the gradient and the risk of losing one's job does not, I turn to the right panel, which switches saving off, so households cannot hold precautionary savings against job loss, leaving every parameter at its calibrated value. Without savings, the unemployment channel also reverses the gradient: on its own it takes the average user to 0.75 times the population average at its calibrated strength, and to 0.27 at 200 percent.

Who uses is decided by a race between two things that change with income: the cost of acquiring drugs and the penalty they inflict. The utility cost of the drug declines with income because a household with a higher income gives up less marginal utility for each dollar it spends on drugs, so buying v units costs a rich household little in terms of utility relative to a poor household.

The penalty inflicted by drugs does not rise with income when it can be partially insured against, but does rise when it cannot be. Unemployment can be partly insured against through saving: a richer household carries more savings into a spell out of work, so the loss is proportional to what it has, and thus no greater in utility terms for a rich household than for a poor one. Death, on the other hand, cannot be insured against. It destroys the whole continuation value, the consumption of a life still to be lived and $\aleph$, the value of being alive at all, so the loss grows with income, deterring high-income households and not low-income ones. Without the ability to save, losing a job takes a household down to $\bar{c}$ whatever it earned, so that loss too grows with income, which is why in the right panel the unemployment channel reverses the gradient as well.

4 The quantitative model

This section describes the model the paper's numbers come from. It keeps the preferences of the stylized model of Section 3 and the two risks a SUD brings there, and adds what that model leaves out: housing,

[10] With both channels at their calibrated strength, as in the calibration, the average user earns 0.74 times the population average, as in the data.

homelessness and vouchers, annual periods, unemployment insurance, income risk during working age, retirement, progressive income taxes, interest on savings, a drug-use decision in every year, recovery, relapse and treatment, SMI, a higher risk of dying, worse employment prospects and a lower chance of recovering while homeless, children who live with their parents, a drug test that misses some users, a balanced government budget and an endogenous drug price. Notation follows Section 3 wherever the two models share an object: θ is the drug weight in both, σ the same risk aversion, b is the same Stone-Geary shifter, $\aleph$ the same flow value of being alive, π_d the same chance that use starts a SUD, $\kappa_{\phi d}$ the same mortality multiplier for the disorder, p_v the same drug price, and β is the same discount factor. The values are in Section 5.

4.1 Demographics and the state and choice sets

At model age $j = 1$, a child becomes an adult, leaves its parent's home, joins the workforce and forms its own household, with a child of its own. The household lives at most J annual periods, works while $j < J_R$ and is retired from J_R, and its child lives with it for the first J_K years, after which the child leaves to form its own household in turn, and so on.

Each period a cohort of unit mass enters at $j = 1$, made up of the children who leave home that period, at their parent's death if that comes before J_K and at J_K otherwise (equation (29)), and replaces the mass of the households who died, such that the mass of households is constant in a stationary equilibrium. The individual state is age j together with

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathbb{R}_+ \times \mathcal{Y}(j) \times \{1,2,3\} \times \{0,1,2\} \times \{0,1\} \times \{0,1,2\}, \tag{9}$$

where a denotes assets, y productivity, e employment status, d the SUD, m SMI and η the household's standing with the voucher programme. The SUD takes three values because the parent's valuation of their child's prospects depends on their history and not only on its present (Section 4.10): $d = 0$ denotes a household that has never had the disorder, $d = 1$ refers to one that had it and has recovered, and $d = 2$ stands for one that has it now. The voucher state is $\eta = 0$, denoting having no voucher, $\eta = 1$, representing holding one, and $\eta = 2$, standing for exclusion from the programme after failing Treatment First's condition, discussed in Section 4.9.

Each period the household chooses the bundle

$$\big(h,\, v,\, \tau,\, a'\big) \in \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathbb{R}_+, \tag{10}$$

where h is the dwelling, of which h_1 is a shelter, v is drug consumption, τ is whether or not a household is in treatment for SUD, available only to a household that has the disorder, and a' is its next period's assets. Labour is inelastically supplied when employed.

4.2 Preferences

Flow utility is the stylized model's, with housing in the bundle and the child present for the first J_K years. Consumption c and housing h are shared through the equivalence scale $\xi(j)$, which scales for two members while the child is at home and for one after, and the drug enters the parent's utility alone, so the parent and the child hold the bundles

$$X_P = \Big[\Big(\frac{c}{\xi(j)}\Big)^{1-\gamma}\Big(\frac{h}{\xi(j)}\Big)^{\gamma}\Big]^{1-\theta}(b+v)^{\theta}, \qquad X_c = \Big[\Big(\frac{c}{\xi(j)}\Big)^{1-\gamma}\Big(\frac{h}{\xi(j)}\Big)^{\gamma}\Big]^{1-\theta} b^{\theta}, \tag{11}$$

and period utility is

$$u\big(c,h,v,\tau\big) \;=\; U(X_P) \;+\; \omega_c\,(\xi(j)-1)\,U(X_c) \;+\; \aleph \;-\; \tau\,\bar{\tau}^{u}, \qquad U(X) = \frac{X^{1-\sigma}}{1-\sigma}. \tag{12}$$

γ is housing's share of the non-drug bundle and θ the drug's share of the whole, σ is the coefficient of relative risk aversion, $\aleph$ is a flow value of being alive, counted once per household, and $\bar{\tau}^{u}$ is the utility cost of a year in treatment, which is described in Section 4.6. The child's term is its own flow utility, which the parent weighs by ω_c, and is counted as $\xi(j) - 1$ of the household; it gets none of the drug, so a parent's drug spending reaches it through a reduction in the consumption and housing they share.

4.3 Income, taxes and transfers

Productivity. Log productivity follows a persistent process over working life, $\log y_{j+1} = \rho_y \log y_j + \zeta_{j+1}$ with $\zeta \sim \mathcal{N}(0, \sigma_\zeta^2)$, and is held fixed at its last working value in retirement. Earnings on the job are $y\,\varepsilon(j)$, with y normalised so that it averages one at entry and $\varepsilon(j)$ the deterministic age profile. A household's productivity at entry is its inheritance from its parent, set out in Section 4.8; its spread combines σ_ζ with a persistent component drawn at entry, and a child entering with the penalty of that section is on a separate block of the grid, with the base grid scaled down for life.

Before-tax income. Before-tax income depends on age and employment,

$$\tilde{y}(j, y, e) = \begin{cases} y\,\varepsilon(j) & j < J_R,\ e = 1 \quad \text{(employed)} \\ \psi_{ui}\,\min\{y\,\varepsilon(j),\ \bar{y}\} & j < J_R,\ e = 2 \quad \text{(insured, first year out of work)} \\ 0 & j < J_R,\ e = 3 \quad \text{(out of work, uninsured)} \\ S(y) & j \geq J_R \quad \text{(retired)}. \end{cases} \tag{13}$$

Unemployment insurance (UI) replaces a share ψ_{ui} of earnings on a base capped at $\bar{y}$, and it is paid only to a share p_{ui} of those losing their job, mimicking the share of insured separations. After the first year, unemployed households, insured and uninsured alike, receive zero UI ($e = 3$). Social security $S(y)$ follows the statutory formula.[11]

The SUD and SMI act on whether the household is employed, through the employment transitions below. There is no disability insurance in the model, so SMI brings no income of its own.[12]

Taxes. Before-tax income is taxed with the Heathcote et al. (2017) function and capital income at a flat rate, so that after-tax income is

$$y^n = \min\left\{\tilde{y},\ \tau_\ell\,\tilde{y}^{\varphi}\right\} \ +\ (1 - \tau_k)\,r\,a, \tag{14}$$

where τ_ℓ scales after-tax income, and so sets the level of the tax, which falls as τ_ℓ rises, and $1 - \varphi$ sets its progressivity; τ_k is the flat tax rate on capital income and r is the exogenous interest rate on savings.[13]

Transfers. Food assistance pays the maximum allotment $\underline{c}$ less the share ϱ of income, and the voucher pays a holder that is eligible this period the rent on the dwelling above a tenant contribution of κ_r of income, up to the fair market rent (*FMR*),

$$\begin{aligned} F &= \max\left\{0,\ \underline{c} - \varrho\,\hat{y}\right\}, \\ Vch &= \mathbf{1}\{\eta = 1\}\,\mathbf{1}\{\hat{y} < \kappa_V y^{\text{med}}\}\,\mathbf{1}\{h \neq h_1\}\,\min\Big\{\max\{0,\ p_r h - \kappa_r \hat{y}\},\ \max\{0,\ FMR - \kappa_r \hat{y}\}\Big\}, \end{aligned} \tag{15}$$

where $\hat{y} = \tilde{y} + ra$ is the income both programmes are means-tested on, κ_V and median income y^{med} are the eligibility rule in Section 4.9, and *FMR* is the average rent paid by housed households in the benchmark. A voucher holder in a shelter keeps the voucher and is paid nothing until it rents again, as long as it keeps its eligibility.

Each entering household receives a one-time endowment T_0 from bequests left by deceased households, and the government collects the rest of the estates of the dead.

[11] Let $y^s = \min\{y, y^{\max}\}$ denote the household's productivity at its last working age up to the taxable maximum $y^{\max}$, the ceiling above which earnings no longer raise the benefit. Then social security payments are $S(y) = \bar{w}\max\left\{\underline{S},\ \alpha_1 \min\{y^s, k_1\} + \alpha_2 \max\{0, \min\{y^s, k_2\} - k_1\} + \alpha_3 \max\{0, y^s - k_2\}\right\}$, where $k_1 < k_2$ are the two bend points, $\alpha_1 > \alpha_2 > \alpha_3$ the shares replaced below, between and above them, $\underline{S}$ the Supplemental Security Income (SSI) payment, and $\bar{w}$ the benchmark's average labour income. Their values are given in Appendix A.

[12] In the data only a minority of adults with SMI receive disability benefits on the basis of a mental disorder: about 3 million, through Social Security Disability Insurance or SSI, against 14.6 million adults with SMI in NSDUH 2023, roughly one in five.

[13] The minimum keeps the labour income tax from turning negative, since at low incomes $\tau_\ell \tilde{y}^{\varphi}$ exceeds $\tilde{y}$ itself.

4.4 The budget constraint

The household's budget constraint is as follows:

$$c + p_v v + a' \;=\; y^n + a + T_0 \mathbf{1}\{j = 1\} \;-\; p_r\, h\, \mathbf{1}\{h \neq h_1\} \;+\; Vch + F, \qquad c \geq F,\; v \geq 0,\; a' \geq 0. \tag{16}$$

Food assistance is paid in kind, so it cannot directly fund the drug, the rent or saving. The drug price is p_v, which is endogenous, whereas the rent per unit of housing p_r is fixed and held at its benchmark value in every simulation.[14] Treatment costs utility only.

4.5 Housing and the homelessness loop

Housing is the four-point grid of (10): h_1 is literal homelessness, sleeping rough or in an emergency shelter, which is free, and $h_2 < h_3 < h_4$ are three rental sizes. Homelessness raises mortality (20), worsens both employment margins (21) and lowers the chance of recovering from the SUD (18), but does not affect SMI (Section 4.7).

4.6 Drug use, the SUD, recovery and treatment

A household that does not have the disorder, either $d = 0$ (never had it) or $d = 1$ (had it but recovered), and uses acquires it ($d = 2$) with probability π_d,

$$\Pr(d' = 2 \mid d \neq 2,\, v) = \pi_d\, \mathbf{1}\{v > 0\}, \tag{17}$$

and a household that has recovered can acquire the disorder again. A household may begin using only up to an age J_E.[15] A household with the SUD recovers, to $d = 1$, with probability

$$\begin{aligned} \pi_r(h, m, \tau) = \min \Big\{ 1,\; \bar{\pi}_r \left(\mathbf{1}\{h = h_1\}\, \mu_H + \mathbf{1}\{h \neq h_1\}\right)\left(\mathbf{1}\{m = 1\}\, \mu_m + \mathbf{1}\{m = 0\}\right) \\ \times \left(\mathbf{1}\{\tau = 1\}\, g + \mathbf{1}\{\tau = 0\}\right) \Big\}, \end{aligned} \tag{18}$$

where $\bar{\pi}_r$ is the baseline annual chance of recovering while housed, without SMI and untreated. It has three multipliers on it: $\mu_H < 1$, which means that recovery is harder in a shelter, $\mu_m < 1$, which means that it is harder with SMI, and $g > 1$, which means that it is more likely in treatment.

Treatment τ is a choice, taken at the utility cost $\bar{\tau}^u$ in (12), and g is what it buys. It does not stop the household using and it does not remove the excess mortality of the disorder while it lasts.

4.7 SMI

m is a two-state Markov chain representing SMI, mental illness severe enough to cause substantial functional impairment. It affects a small share of adults, whereas any mental illness affects roughly a quarter of adults and includes mild anxiety. Onset follows an age profile $\varsigma_m(j)$, normalised to one at age 26 and scaled by the level ω_m, and a household that carries the SUD is $\iota > 1$ times more likely to develop SMI than one that does not,

$$\Pr(m' = 1 \mid m = 0, d) = \min\left\{1,\; \omega_m\, \varsigma_m(j)\left(\mathbf{1}\{d = 2\}\, \iota + \mathbf{1}\{d \neq 2\}\right)\right\}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m. \tag{19}$$

A household with SMI recovers from it with the probability λ_m each year, whatever its housing and whether or not it carries the SUD.

[14] In earlier versions of this work, I solved the model with endogenous labour supply, rents, homeownership and house prices. These margins turned out to have a negligible effect on the results, so for conciseness I removed them and hold the rent fixed.

[15] Without it there is little-to-no consequence at old ages to using: too little life remains for the disorder to damage, so the model produces too much of it. The cause is that death is certain at the last age J, so close to it a household knows how little life remains, whereas in real life the vast majority of people do not know exactly when they will die. J_E is the equivalent of age 80 when the model runs to age 85. It stops only households without a SUD from starting to use; a household that already has a SUD at that age keeps using.

SMI then affects three objects: the chance of finding a job when unemployed, through (21), mortality, through (20), both in Section 4.8, and the SUD recovery hazard, through μ_m in (18) in Section 4.6. It does not enter the utility function.

The SMI chain does not depend on housing, because in the At Home / Chez Soi randomised trial housing had no detectable effect on the symptoms of homeless adults with SMI (Stergiopoulos et al., 2015).

4.8 What the SUD, SMI, and homelessness cost

Death risk. Write $1-\bar{\phi}(j)$ for the probability of dying at age j of a household that is housed and has neither the disorder nor SMI; the three multipliers below raise it for every other household. Write $q^{\,0}(j,h) = \min\{1, \left(\mathbf{1}\{h=h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h\neq h_1\}\right)(1-\bar{\phi}(j))\}$ and $q^{\,1}(j,h) = \min\{1, \kappa_{\phi d}\left(\mathbf{1}\{h=h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h\neq h_1\}\right)(1-\bar{\phi}(j))\}$ for the probability of dying without and with the disorder, respectively. The probability of dying is then

$$1-\phi(j,h,d,m) = \min\Big\{1,\ \big(\mathbf{1}\{m=1\}\,\kappa_{\phi m} + \mathbf{1}\{m=0\}\big)\,q^{\,0}(j,h) + \mathbf{1}\{d=2\}\big[q^{\,1}(j,h) - q^{\,0}(j,h)\big]\Big\}, \tag{20}$$

so $\kappa_{\phi H} > 1$ multiplies the death probability when a household is in a shelter, $\kappa_{\phi d} > 1$ when a household has a SUD, and $\kappa_{\phi m} > 1$ the baseline risk when a household has SMI, to which the excess risk of the disorder is added.

Unemployment risk. A job is lost with probability π^{eu} and found with probability π^{ue},

$$\begin{aligned} \pi^{eu}(y,h) &= \min\Big\{1,\ \bar{\pi}^{eu}\,\varpi(y)\,\big(1+\mathbf{1}\{h=h_1\}\,\kappa_H\big)\Big\}, \\ \pi^{ue}(h,d',m') &= \min\Big\{1,\ \frac{\bar{\pi}^{ue}\,\big(1-\mathbf{1}\{h=h_1\}\,\kappa_H\big)}{\big(\mathbf{1}\{d'=2\}\,\kappa_d^{ue} + \mathbf{1}\{d'\neq 2\}\big)\big(\mathbf{1}\{m'=1\}\,\kappa_m^{ue} + \mathbf{1}\{m'=0\}\big)}\Big\}, \end{aligned} \tag{21}$$

with $\bar{\pi}^{eu}$ and $\bar{\pi}^{ue}$ the baseline separation and job finding rates, $\varpi(y)$ a factor that makes separations likelier at low productivity, and the divisors κ_d^{ue} and κ_m^{ue} making one harder to find with the SUD and with SMI. Neither the SUD nor SMI raises separations; each only reduces the chance of regaining employment. κ_H does both for a household in a shelter.

The child's productivity penalty. A child whose parent had the disorder at any point while the child lived at home, before J_K or before the parent died if that came first, starts adult life with its productivity lowered by the proportion δ_d in (22) below, and the penalty stays with the child for life.

A child's log productivity at entry is drawn around its parent's, with persistence ρ. The innovation keeps the spread of productivity at entry the same in every generation. If the parent had the disorder at any point before the child became an adult, which is at the parent's death if that comes before J_K and at J_K otherwise, the penalty scales the child's productivity down for life:

$$\begin{aligned} \log y_1^{\text{child}} \;&= \rho\,\log y^{\text{parent}} + (1-\rho)\,\mu + \epsilon \\ &\quad + \log(1-\delta_d)\,\mathbf{1}\{\text{parent has } d\geq 1 \text{ when its child becomes an adult}\}, \end{aligned} \tag{22}$$

with y_1^{child} the child's productivity in the period it becomes an adult in the workforce ($j=1$), y^{parent} the parent's productivity in the period just before its child becomes an adult, μ the mean of log productivity at entry and ϵ normally distributed with mean zero and variance $1-\rho^2$ times the variance of log productivity at entry. Because the penalised child's own productivity is lower, its own child inherits a lower draw too, so, through the persistence, the penalty ripples through the whole lineage.

4.9 The voucher and the two policies

A household is eligible for a voucher while its income is below the share κ_V of median income,

$$\hat{y} \;<\; \kappa_V\, y^{\text{med}}, \tag{23}$$

with $\hat{y}$ the income of (15). A holder keeps its voucher while it is eligible, and an eligible household without one draws one with probability π_V. Under Housing First that is the whole rule.

Under Treatment First the programme is conditional. A household fails the condition if it uses or has the disorder, $v > 0$ or $d = 2$, whether or not it is in treatment. Income-eligible holders and income-eligible households that draw a voucher are tested, at a cost c_{test} each to the government, and one that fails is detected with probability p_{det}. This means that enforcement is not perfect: with probability $1 - p_{\text{det}} > 0$ a household that fails the condition actually gets to keep its voucher. A detected holder keeps this period's payment and loses the voucher from the next period; a detected applicant never receives it. In both cases, the household is excluded from the voucher programme for a period of time ($\eta = 2$): it cannot draw a voucher whatever its income, and it returns to the lottery ($\eta = 0$) with probability ψ_x each year.

None of p_{det}, ψ_x and c_{test} binds in the benchmark, where no voucher is conditional, so all three are set with the experiment in Section 6 rather than in the calibration.

4.10 The household's dynamic problem

Write $\Gamma\big(s' \mid s,\, h, v, \tau, a'\big)$ for the law of motion of the state that the transitions above define, given the state and the choices. Then, for $j = 1, \ldots, J$ and with $V(J{+}1, \cdot) = 0$,

$$
\begin{aligned}
V(j,s) = \max_{\{h,\,v,\,\tau,\,a'\}} \Bigg\{ & U(X_P) + \omega_c(\xi(j) - 1)\, U(X_c) + \aleph - \tau\bar{\tau}^{u} \\
& +\ \beta\, \phi(j,h,d,m) \int_{\mathcal{S}} V(j{+}1, s')\, \Gamma\big(\mathrm{d}s' \mid s, h, v, \tau, a'\big) \\
& -\ \beta\, \chi(j)\, \omega_c\, D(y,d) \Bigg\}, \qquad \chi(j) = \begin{cases} 1 - \phi(j,h,d,m) & j < J_K \\ 1 & j = J_K \\ 0 & j > J_K, \end{cases}
\end{aligned}
\tag{24}
$$

subject to the budget (16), the income and transfer rules (13) to (15) in Section 4.3, and the restrictions that treatment is available only to a household with $d = 2$ and that no household begins using after J_E.

The first line is the flow of (12). The second is the household's own continuation. The third is the impact of the parent on the child's adult value. The latter is counted only when the child leaves: at J_K with certainty, and before then with the probability that the parent dies first. $D(y,d)$ is what the parent's disorder costs the child, in the child's own lifetime value,

$$
D(y,d) = \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ 0\big] - \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ d\big] \ \geq\ 0, \tag{25}
$$

where the child is valued entering at $j = 1$ with its initial states, described in Section 4.13, and its productivity is drawn as in (22) in Section 4.8 from a parent with productivity y. In the first term of (25) the parent has never had the disorder, and in the second they have their actual d in the period the child leaves. If they never had a SUD, there is no charge to the parent ($D(y,0) = 0$). However, if they had a SUD before the child leaves ($d \geq 1$), they are charged ($D(y,d) > 0$ for $d \geq 1$).[16]

One weight ω_c multiplies both child terms, the flow of childhood and the value of the child's adulthood. At $\omega_c = 0$ the damage a household does to its child is entirely external; at $\omega_c > 0$ it internalises the share ω_c and only $1 - \omega_c$ remains external.

The policy functions $h(j,s)$, $v(j,s)$, $\tau(j,s)$ and $a'(j,s)$ are the dwelling, drug use, treatment and saving that solve (24) at age j and state s.

[16] The term is a difference and not a level. Written as a level, ω_c times the child's value, it would be worth more to a household the likelier it is to die, because it is the death branch that carries it, so it would pay most to the households with a SUD, for reasons that have nothing to do with the child. As a difference, the term is zero for a parent who has never had the disorder and a cost otherwise.

4.11 The government

Write $\Phi(j,s)$ for the distribution of the households of age j over the states s, whose law of motion is in Section 4.13. The government's budget constraint is

$$\begin{aligned}\mathcal{R} \;=\; \sum_{j=1}^{J} \int_{\mathcal{S}} \Big[\tilde{y}(j,y,e) - \min\big\{\tilde{y}(j,y,e),\, \tau_\ell\, \tilde{y}(j,y,e)^{\varphi}\big\} + \tau_k\, r\, a \\ + \big(1-\phi\big(j,h(j,s),d,m\big)\big)\, a'(j,s) \Big]\, \mathrm{d}\Phi(j,s) \;-\; T_0 \\ - \sum_{j=1}^{J} \int_{\mathcal{S}} \Big[UI(j,y,e) + S(j,y) + F(j,s) + Vch(j,s) + c_{\text{test}}\, \mathbf{1}^{\text{test}}(j,s) \Big]\, \mathrm{d}\Phi(j,s).\end{aligned} \tag{26}$$

The first line is revenue from the labour and capital income taxes of (14). The second is the estates of the households who die, which the government collects, less the endowment T_0 it gives the unit mass of entrants. The third is what the government pays: unemployment insurance, $UI(j,y,e) = \tilde{y}(j,y,e)\,\mathbf{1}\{j < J_R,\, e=2\}$; social security, $S(j,y) = S(y)\,\mathbf{1}\{j \geq J_R\}$; food assistance $F(j,s)$ and the voucher $Vch(j,s)$ of (15), at the household's dwelling $h(j,s)$; and, under Treatment First, the cost c_{test} of each household tested, $\mathbf{1}^{\text{test}}(j,s) = 1$.

In the benchmark the tax parameters are set as in Section 5 and $\mathcal{R}$ is whatever surplus they produce. In every experiment the taxes are held at their benchmark values and, unless the experiment holds it, the probability of drawing a voucher π_V adjusts so that $\mathcal{R}$ equals its benchmark value $\bar{\mathcal{R}}$.

4.12 The drug market

The price of the drug clears its market,

$$p_v = \nu_1 S_v^{1/\varepsilon_s}, \qquad S_v = \sum_{j=1}^{J} \int v(j,s)\, \mathrm{d}\Phi(j,s), \tag{27}$$

where ε_s is the price elasticity of supply and ν_1 is normalised so that the price is one in the benchmark, and then held fixed across experiments.

4.13 The law of motion

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi(j{+}1, \mathcal{S}') \;=\; \int_{\mathcal{S}} \phi\big(j,h(j,s),d,m\big)\, \Gamma\big(\mathcal{S}' \mid s,\, h(j,s), v(j,s), \tau(j,s), a'(j,s)\big)\, \mathrm{d}\Phi(j,s), \tag{28}$$

for $j = 1, \ldots, J-1$ and every measurable set $\mathcal{S}'$ of the state space $\mathcal{S}$ of (9), with the integral over the state s of the households of age j, together with the inflow of entrants

$$\Phi(1, \mathcal{S}') \;=\; \frac{\sum_{j \leq J_K} \int_{\mathcal{S}} \chi(j)\, \mathcal{N}\big(\mathcal{S}' \mid s\big)\, \mathrm{d}\Phi(j,s)}{\sum_{j \leq J_K} \int_{\mathcal{S}} \chi(j)\, \mathrm{d}\Phi(j,s)}, \tag{29}$$

where the integrals are again over s, $\mathcal{N}$ places the child according to (22) and $\chi(j)$ is the weight defined in Section 4.10. An entrant starts with the endowment T_0, is employed, has never had the disorder, holds no voucher, and carries SMI with the prevalence observed at age 26.[17]

[17] A disorder acquired before that age is therefore acquired in the model only after entry; however, the share of adults with a SUD still matches the data (Section 5).

4.14 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\left\{\{V_t(j,\cdot)\}_{j=1}^{J}\right\}_{t=0}^{\infty}$;
- policy functions $\left\{\{h_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\right\}_{t=0}^{\infty}$;
- distributions $\left\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\right\}_{t=0}^{\infty}$; and
- prices, policies and aggregates $\left\{p_{vt},\, \pi_{Vt}\right\}_{t=0}^{\infty}$, together with $\bar{y}$, y^{med} and FMR, which carry no date because they are kept fixed at their benchmark values,

such that, at every date t,

1. the policy functions solve (24) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;
2. $\Phi_{t+1}(j{+}1,\cdot)$ follows from $\Phi_t(j,\cdot)$ by the law of motion (28) together with the entrants of (29) and (22);
3. the government budget (26) holds at π_{Vt} with $\mathcal{R} = \bar{\mathcal{R}}$;
4. the drug market clears at p_{vt}, so (27) holds.

A stationary equilibrium is the special case in which all of these are constant and $\Phi_{t+1} = \Phi_t$, which is what the calibration and the long-run policy comparisons report. The sequence version is what the transition between them solves.

5 Calibrating the quantitative model

The model has fifty-eight parameters, some of which vary by age: Thirty-seven are calibrated externally, from data or the literature, and twenty-one internally. The twenty-five that carry this paper's story are in Table 3, with the moment, source, data and model value behind each, and are discussed below in numbered blocks of parameters that work through similar mechanisms. The other thirty-three, common to most life-cycle models, are in Appendix A to save the reader time; its blocks continue the numbering. Three further parameters govern drug testing and exclusion from the voucher; they play no role in the benchmark and are set with the experiment in Section 6. `data/DATA_APPENDIX.md` in the replication package documents the dataset, year, universe and computation behind every estimate, and the package carries all datasets, so nothing has to be downloaded.

5.1 The twenty-five parameters that carry the story of this paper

Block 1: Who uses drugs. The five parameters in this block set participation and, more importantly, *who* participates. Neither policy acts on them directly; they decide how much there is for a policy to act on.

θ, the weight on drugs in utility. It is the drug's weight in the bundle, so it mostly sets how much a household that uses spends on drugs, but a higher weight also makes using at all worth more. As it rises, use therefore rises on both the extensive and the intensive margin, and with it the externalities and what either policy has to act on. It is calibrated internally, at 0.17, to aggregate drug spending by users over their aggregate income, 16.0 percent, from retail drug spending in Midgette et al. (2019) and the incomes of adults aged 26 and over who use in NSDUH 2024.

b, the marginal utility from the first unit of drugs. The marginal value of the first unit of drugs goes as θ/b, so a larger b makes the first units worth less and a household with little to spend buys none. As b falls, the marginal utility from the first unit rises, and so do overall drug use and the externalities that policy can affect. It targets a participation rate of 8.90 percent, the NSDUH 2024 rate of a drug use disorder of any severity among adults aged 26 and over.[18]

[18] A user in the model is therefore a household whose use amounts to a disorder, mild included, of any illicit drug, marijuana among them, and a household with a SUD is one whose disorder is moderate or severe (next item). Use that never reaches a disorder is not modelled.

Table 3: The twenty-five parameters that carry the story of this paper

Parameter	Description	Set	Moment and source	Data	Model	Value
Block 1: Who uses drugs						
θ	Weight on drugs in utility	Int.	Drug spending over income, Midgette et al. (2019)	16.0%	14.8%	0.165
b	Marginal utility from the first unit of drugs	Int.	Any drug use disorder, NSDUH 2024	8.90%	8.96%	0.0054
π_d	Chance that a year of use leads to a SUD	Int.	Share of users with a SUD, NSDUH 2024	41.3%	42.6%	0.0345
$\kappa_{\phi d}$	Increased mortality with a SUD	Int.	Income of drug users over average, NSDUH 2024	74%	62%	4.00
$\aleph$	Flow value of being alive	Int.	VSL, Viscusi and Aldy (2003)	80	81	87.6
Block 2: Costs of homelessness and vouchers						
κ_d^{ue}	How much less likely it is to regain employment with a SUD	Int.	SUD share of the homeless, Gutwinski et al. (2021)	21.7%	20.2%	3.78
h_1	Service flow of homelessness	Int.	Homelessness band, HUD PiT 2023 and Burt et al. (2001)	0.40–1.00%	0.62%	0.0216
$\kappa_{\phi H}$	Mortality when homeless	Ext.	Meyer et al. (2023)	1.60	n.a.	1.60
κ_H	Employment penalty of homelessness	Ext.	Desmond and Gershenson (2016)	0.20	n.a.	0.20
$\underline{c}$	Food stamp maximum allotment	Int.	SNAP outlays over output	0.42%	0.43%	0.0879
π_V	Chance of drawing a voucher	Int.	Recipients over eligible, HUD 2023	26.6%	27.5%	0.0675
T_0	Endowment at entry to adulthood ($j = 1$)	Int.	Share of homeless aged 25 to 34, HUD PiT 2024	26.0%	26.4%	0.111
Block 3: SMI chain						
ι	SUD multiplier on SMI onset	Ext.	Danish registers, Nielsen et al. (2017)	2.00	n.a.	2.00
λ_m	Annual chance of recovering from SMI	Ext.	SMI ten-year recovery rate, Åsbø et al. (2022)	50%	n.a.	0.066
ω_m	Level of the SMI onset profile	Int.	SMI share of the population, NSDUH 2024	4.82%	4.83%	0.0037
κ_m^{ue}	How much less likely it is to regain employment with SMI	Int.	SMI share of the homeless, Gutwinski et al. (2021)	25%	24.7%	7.71
μ_m	SUD recovery multiplier for SMI	Ext.	Dual diagnosis relapse, Bradizza et al. (2006)	0.82	n.a.	0.82
$\kappa_{\phi m}$	Mortality multiplier for SMI	Ext.	Olfson et al. (2015)	2.50	n.a.	2.50
Block 4: Recovery and treatment						
$\bar{\pi}_r$	Baseline SUD recovery rate	Ext.	SUD ten-year remission rate, Vaillant (1995)	22%	n.a.	0.025
μ_H	SUD recovery when homeless	Int.	Abstinence, housed/homeless, Milby et al. (2005)	1.47	1.45	0.696
g	Treatment effect on recovery	Ext.	Remission on buprenorphine, Paul et al. (2024)	2.00	n.a.	2.00
$\bar{\tau}^u$	Utility cost of treatment	Int.	Treatment take-up, NSDUH 2024	24.5%	24.9%	4.49
Block 5: Externalities						
ω_c	Parent's weight on its child	Int.	Two-adult over lone-parent households' use, less one, NSDUH 2024	17.5%	17.7%	0.085
δ_d	Earnings the child loses because of a parent's SUD	Ext.	Earnings lost in utero, Nilsson (2017)	0.24	n.a.	0.24
Block 6: Drug market						
ε_s	Price elasticity of drug supply	Ext.	Mace et al. (2020)	1.46	n.a.	1.46

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

π_d, the chance that a year of use leads to a SUD. This is what makes using *risky*: the step from drug use to a disorder. As it rises, the externalities rise, and the moral hazard of Housing First turns an extra voucher into more SUD, which Treatment First then aims to prevent. It targets the share of users who have a SUD, 41.3 percent among adults aged 26 and over in NSDUH 2024.

$\kappa_{\phi d}$, increased mortality with a SUD. As the stylized model of Section 3 shows, this is a key parameter. Mortality truncates a user's value of life, and as it rises it hits the households with the highest values of life hardest, because it is an uninsurable risk, which reverses the income gradient. It is calibrated to the average income of users, at 4.00, within the range of the mortality rate ratios of a SUD in the literature: 3.91 in the Danish registers (Plana-Ripoll et al., 2019) and 6.6 among Norwegians in specialist care for one (Heiberg et al., 2018).

$\aleph$, the flow value of being alive. $\aleph$ is the other half of the mortality mechanism: $\kappa_{\phi d}$ sets how much a SUD raises the chance of dying, and $\aleph$ what the life lost is worth. As it rises, dying costs more and use falls. It targets a VSL of 80 times average labour income (Viscusi and Aldy, 2003).[19]

Block 2: Costs of homelessness and vouchers. Both policies work on this block, from opposite ends: each parameter says how welfare reducing a state is, and so what a voucher and its removal are worth in it.

κ_d^{ue}, how much less likely it is to regain employment with a SUD. An unemployed household with a SUD finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_d^{ue} > 1$. As κ_d^{ue} rises, the expected unemployment spells of households with a SUD lengthen and they are likelier to lose their housing: this is the link from the disorder to the street, and it shapes how much Treatment First deters. It is calibrated internally to the SUD share of the homeless, 21.7 percent in Gutwinski et al. (2021).

h_1, the service flow of homelessness. This parameter sets how welfare reducing the outside option is, which affects the value of life: a household with little to lose from dying faces a smaller mortality cost of using. As h_1 rises, a shelter is less welfare reducing, a voucher is worth less and Treatment First has less to act on. It targets a homelessness rate within 0.40 to 1.00 percent of households: the floor

[19] No household has a negative value of life in equilibrium, so death is never optimal. Because the value of being alive enters utility additively and utility is concave, the VSL rises faster than consumption, roughly with its square at the model's risk aversion, as in Hall and Jones (2007). Estimates of how it rises with income are lower across people at a point in time, 0.5 to 0.6 in Viscusi and Aldy (2003), and higher over time, 1.5 to 1.7 in Costa and Kahn (2004).

is HUD's 2023 PiT count, which counts a single night; the ceiling is the share homeless at some point over a year, from Burt et al. (2001). Its value is 53 percent of the smallest rental (Appendix A), and the model delivers an aggregate homelessness rate of 0.62 percent.

$\kappa_{\phi H}$**, mortality when homeless.** This parameter sets how much more likely a homeless household is to die. As it rises, the cost of homelessness rises, a voucher is worth more and Treatment First's deterrence is stronger. It is set externally to multiply baseline mortality by 1.60 at every age, the mortality ratio for homeless adults against poor housed adults in Meyer et al. (2023).

κ_H**, the employment penalty of homelessness.** This parameter sets how much likelier a household in a shelter is to lose its job and how much less likely it is to find one: separations are $1 + \kappa_H$ times as frequent and job finding $1 - \kappa_H$ times as fast. Housing First recovers that employment directly, whereas Treatment First does so only as the fall in SUD lowers unemployment and homelessness over time. As it rises, homelessness costs more, and both policies have more to act on. It is set externally at 0.20, from Desmond and Gershenson (2016).

$\underline{c}$**, the food stamp maximum allotment.** This parameter sets how low consumption can go for the poorest households, with no income, assets or voucher, and so where utility is bounded from below. As it rises, the value of being alive rises, mortality costs more and Treatment First has more to deter. It is calibrated internally to food stamp spending over output, 0.42 percent: outlays from the United States Department of Agriculture (USDA) over GDP.

π_V**, the chance of drawing a voucher.** This is the policy dial itself, and it holds the whole trade-off. Raising it houses more households, which reduces the *childhood homelessness externality*, and makes life worth more, which makes Treatment First's deterrence more effective. In the benchmark it is calibrated internally to 0.068, so that the share of income-eligible households who hold a voucher is aligned with the 26.6 percent in HUD's Picture of Subsidized Households (PSH), 2023.

T_0**, the endowment at entry to adulthood ($j = 1$).** A child who becomes an adult starts adult life with T_0 times average earnings, which sets how likely a young adult is to end up in a shelter in its first years. As it rises, fewer young adults become homeless, and Treatment First affects behaviour less in the child-rearing years, when parents' choices affect their children. Its moment is the share of the homeless aged 25 to 34, 26.0 percent in HUD's 2024 PiT count.

Block 3: SMI chain. This block sets how much further Treatment First reaches beyond the disorder it prevents: a SUD raises the chance of SMI, and SMI lowers employment and raises mortality.

ι**, the SUD multiplier on SMI onset.** The hinge of the block: without it, SMI is exogenous to everything a policy does. As it rises, a SUD causes more SMI, and Treatment First's return rises, because preventing the disorder now prevents some SMI too. It is set externally at 2.00, between the hazard ratio of 1.34 for any SUD that Hjorthøj et al. (2018) estimate for the conversion from schizotypal disorder to schizophrenia and the 6.04 for any substance abuse that Nielsen et al. (2017) estimate for the onset of schizophrenia, both in the Danish registers.

λ_m**, the annual chance of recovering from SMI.** This parameter is the annual hazard of recovering from SMI, and so sets how long an episode is expected to last. As it rises, SMI is less persistent, and Treatment First's deterrence matters less, since the SMI spells a SUD can set off are shorter on average. It is set externally at 0.066, the annual hazard that gives a 50 percent chance of recovery within ten years, the ten-year clinical recovery rate for bipolar disorder in the first-episode psychosis cohort of Åsbø et al. (2022).

ω_m**, the level of the SMI onset profile.** This parameter scales the chance of developing SMI at every age, and so how many households have it. As it rises, a SUD's effect on SMI is larger, and Treatment First has more to act on. It is calibrated internally to the share of the population with SMI, 4.82 percent, the prevalence in NSDUH 2024.

κ_m^{ue}**, how much less likely it is to regain employment with SMI.** The same as κ_d^{ue} for SMI: a household with SMI finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_m^{ue} > 1$. As it rises, expected unemployment spells with SMI lengthen and SMI costs more. Its moment is the SMI share of the homeless, 25 percent in Gutwinski et al. (2021), since a household out of work longer is likelier to become homeless.

μ_m, **the SUD recovery multiplier for SMI.** The chain's second direction: a SUD raises the chance of SMI, and SMI makes a SUD more persistent. As it falls, recovering from a SUD is less likely, and Treatment First has more to overcome. It is set externally at 0.82 from the dual-diagnosis relapse rates in Bradizza et al. (2006).

$\kappa_{\phi m}$, **the mortality multiplier for SMI.** The chain's last link, through which it shortens life. As it rises, more households with SMI die, and Treatment First has more to work on, because less drug use means less SUD, less SMI and fewer deaths. It is set externally at 2.50, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

Block 4: Recovery and treatment. Both policies work through this block: Treatment First encourages seeking treatment, and Housing First buys recovery by housing people, since recovery is likelier while housed. The first two parameters set how much recovery there is to buy, the last two how much of it a policy can reach.

$\bar{\pi}_r$, **the baseline SUD recovery rate.** This parameter is the annual chance that a housed household recovers from a SUD, the level everything else in the block scales. As it rises, disorders are expected to end more quickly on their own, and there is less for either policy to do. It is set externally at 0.025, the annual hazard implied by the ten-year remission rate in Vaillant (1995).

μ_H, **SUD recovery when homeless.** This parameter sets how much less likely recovery is in a shelter, and so how much recovery a voucher buys beyond shelter. As it falls, housing buys more recovery and Housing First is more effective in curing a SUD; near one, Housing First has little effect. It targets the effect of housing on recovery in Milby et al. (2005), who find that housing improves recovery chances by 47.3 percent.

g, **the treatment effect on recovery.** The instrument through which Treatment First reaches households that already have a SUD, which matters most in the short run. As it rises, treatment is likelier to end a SUD, and Treatment First has a larger margin to act on. It is set externally at 2.00, which is how much medication for opioid use disorder raises remission: rates near double those without it (Paul et al., 2024), and randomised trials of agonist medication against placebo find at least as large an effect on abstinence (Fudala et al., 2003).

$\bar{\tau}^u$, **the utility cost of treatment.** Take-up is a choice, and this utility cost is what makes households decline a treatment that would raise their chance of recovering. As it rises, fewer take treatment up, and Treatment First induces fewer households with a SUD to seek it. It is calibrated internally to treatment take-up: 24.5 percent of adults with a SUD received substance use treatment in the past year in NSDUH 2024.

Block 5: Externalities. These two parameters decide what is an externality and how large it is.

ω_c, **the parent's weight on their child.** A parent bears the share ω_c of what its choices cost its child, in childhood and through the productivity penalty. As it rises, the parent counts more of the child's loss, so the *budget, earnings and childhood homelessness externalities* left for either policy to correct are smaller. It is identified by a counterfactual moment: with every other parameter unchanged, drug use with the child given no weight against drug use at ω_c.[20] The data moment holds the household fixed as well. In NSDUH 2024, an unmarried adult living with one other adult and no child was 17.5 percent more likely to have used illicit drugs in the past year than a lone parent living with one child, with the lone parents reweighted to the other group's age, sex, kind of unmarried (never married, divorced or separated, or widowed) and family income; with two members in each household, equal family income is equal income per person.[21]

[20] All the results continue to hold with $\omega_c = 0.69$, the weight on children that Nishiyama (2000) estimates, which is at the upper end of the estimates in the literature. Quantitative models in the dynastic tradition of Barro and Becker (1989) typically use lower weights (Lee and Seshadri, 2019; Abbott et al., 2019).

[21] This moment counts any past-year use of an illicit drug, a broader measure than the model's, because too few lone parents in the sample have a disorder for the comparison to give a statistically reliable measure. It is therefore used only for how much use responds, in proportion, and not for its level: the model matches it as the proportional rise in its own users.

δ_d, **earnings the child loses because of a parent's SUD.** The child inherits not the disorder but a worse productivity draw, which makes it likelier to be poor, to use, to acquire a SUD and SMI, to be homeless and to die earlier. As it rises, with $\omega_c < 1$, the *earnings externality* grows, which favours whichever policy prevents more drug use: Treatment First by deterring the parent, Housing First by making recovery likelier once housed. It is set externally at 0.24, the earnings loss Nilsson (2017) measures for children exposed in utero to a Swedish alcohol experiment.[22]

Block 6: Drug market. One parameter makes the drug price endogenous, so that it moves with demand.

ε_s, **the price elasticity of drug supply.** This is a general-equilibrium offset that runs against Treatment First: deterring users lowers demand and the price, and a cheaper drug draws other households in. As it rises, supply is more elastic and the price moves less, so the offset shrinks. I set it to 1.46, the estimate of Mace et al. (2020) for cannabis on Washington State administrative data.[23]

5.2 Validation

Table 4 compares the model with moments it was not fitted to, in numbered blocks, each with why it matters for the experiment.

Block 1: Homelessness. The first row gives each age band's share of the homeless; the youngest band is the target of T_0, the other four are free. The model puts 37.9 percent of the homeless at ages 35 to 44, against 27.2 in HUD's PiT count, and 11.4 percent at 55 and over, against 25.9. Of the homeless in the model, 21.4 percent are employed, against 46 in the data, and their income is 6.5 percent of the average, against 19.1 in Meyer et al. (2024). As in the data, most of the homeless are of child-rearing age and poor, so the model places Treatment First's margin, the voucher a household holds or loses, at the ages where it reaches children.

Block 2: Who uses and who has a SUD. Users in the model are somewhat younger than in the data: 67 percent of them are 26 to 34, against 45 in NSDUH 2023, and their mean age is 34.6, against 39.6. Unemployment among households with a SUD is 14.5 percent, against 16.2 in the data. Households with SMI are 2.64 times more likely than others to use drugs, against 2.44, and excess deaths due to SUD account for 14.5 percent of deaths at ages 25 to 44, against 20.0 percent from drug poisonings in the Centers for Disease Control and Prevention (CDC) WONDER database. Both policies reach a child through its parent's decisions, and users are mostly of child-rearing age, in the model as in the data. And since households with SMI use drugs and are homeless at higher rates than the rest, as in the data, the model also captures that Treatment First's screen falls disproportionately on them, since they are the likeliest to fail it.

Block 3: Housing and vouchers. In the model, 58.7 percent of housed households spend over thirty percent of their income on rent, against 43.2 percent of renters in the SCF. A voucher holder pays 39.0 percent of its income in rent,[24] against 31.2 in HUD's PSH, and 39.2 percent of voucher holders live mainly on wages, against 28.0; the others are mostly retired or unemployed. Housing takes a large share of poor households' income, so the consumption Treatment First puts at stake when it conditions the voucher is of a similar size in the model as in the data.

[22] The penalty carries only 11 percent of the welfare gain from Treatment First (Section 6.2), so the ranking of the two policies does not rest on its size.

[23] To the best of my knowledge, there is no direct estimate for an illicit drug, and cannabis is the closest market there is: it remains a Schedule I substance federally, so even a state-licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces.

[24] The rule sets the tenant's contribution at thirty percent of its income, but the voucher pays only up to the fair market rent, so a holder that rents a dwelling above it pays the difference itself, which raises its share above thirty percent, as it does in HUD's data.

Block 4: Income and wealth distributions. The lowest-income fifth of households receive 3.40 percent of income in the model, against 2.71 in the SCF, and hold 0.27 percent of net worth, against −0.23; the bottom half holds 6.3 percent, against 2.2. Both policies act on the households that receive vouchers, the poorest, and the model captures their relative poverty broadly in line with the data.

Block 5: Drug price elasticities. The calibration targets neither the participation nor the intensive elasticity of drug demand. The model's participation elasticity is −1.32, somewhat larger in absolute value than the estimates of about −0.3 to −0.9 in Saffer and Chaloupka (1999) and Grossman and Chaloupka (1998), and its intensive elasticity, over all users, is −0.63, in the middle of the estimates of about −0.3 to −0.8 in Grossman and Chaloupka (1998) and Olmstead et al. (2015). These elasticities are the response of drug use to its price, which is separate from the response to Treatment First itself. The latter, holding the drug price fixed, is in line with the evidence in the short and the long run (Block 8 below). Given that, the higher price response can only work against Treatment First: as Treatment First lowers the drug price, more households return to using, and the more elastic participation is, the more of them do. A welfare gain from Treatment First is therefore conservative in this respect.

Block 6: Transmission. A parent's SUD and homelessness also carry over into its child's adult life, in the model as in the data. A child of a parent with a SUD has 1.46 times the odds of acquiring one in the model, within the range of 1.44 to 2.51 in Mellentin et al. (2016) and Yoon et al. (2013). An adult who is homeless has 6.23 times the odds of having had a homeless parent, above the 1.8 to 2.4 of Bassuk et al. (1997).[25] Transmission in the model is therefore at the low end of the evidence on the SUD, the margin Treatment First acts on, and above it on homelessness. Since Treatment First leaves the share of children in a shelter roughly constant, the excess transmission of homelessness favours neither policy, while transmission of the SUD at the low end of the evidence understates what Treatment First does for children, so if it still raises welfare in the experiments below, that result is conservative. Furthermore, in the model, 5.81 percent of children live with a parent who has a SUD; in NSDUH 2024, 4.6 percent of children whose parents are aged 26 or older live with a parent who has a moderate or severe drug use disorder, implying that the externalities on the child captured by the model reach a share of children similar to that in the data.

Block 7: Losing a voucher. Enrolling in Housing First was estimated to lower the three-year mortality of homeless veterans with mental illness by 4.6 percentage points (Costantini, 2025). In the model, with the drug price and π_V held, I compare households with SMI whom Treatment First's revocation puts on the street with households with SMI who keep their voucher: losing it raises three-year mortality by about 3.6 percentage points, the same order of magnitude. A higher risk of death is the main cost Treatment First imposes on those it screens out, and the model does not understate it.

Block 8: Deterrence. The closest evidence I am aware of on how much a drug test with a severe penalty deters use is the US military's random testing (Section 2): among service members who stayed after it began testing at random and dismissing those who failed, past-year marijuana use fell by about 61 percent of what it would otherwise have been (Mehay and Pacula, 1999). The model cannot reproduce that comparison exactly. Its nearest counterpart is this: I apply Treatment First to the benchmark's households, holding the distribution, the drug price and π_V at their Housing First levels, and follow each user subject to the screen: every voucher holder and every eligible household offered a voucher. Of those without a SUD, 59 percent stop when the screen is introduced[26], and none starts. Those with a SUD, 67 percent of the users subject to the screen, cannot stop on impact, since a disorder ends only with recovery; counting them, 19 percent stop. Deterrence is how Treatment First reduces drug use and through it the externalities on children.[27] Their

[25] Bassuk et al. (1997) compare 220 homeless mothers with 216 low-income housed mothers, so both groups are poor and the odds ratios, 1.8 for a homeless female caretaker and 2.4 for a male one, separate the transmission of homelessness from the poverty around it; a ratio against the general population would credit the housing penalty with everything else that differs between poor children.

[26] In the long run, with the drug price and π_V both held at their Housing First levels (fifth column of Table 5), drug use falls by 64 percent, a similar fall.

[27] The military figure is for any past-year use of marijuana, a broader measure than the model's, whose users have a disorder of any drug, marijuana among them. Therefore, the two are compared as proportional responses. Counting the users with a

Table 4: Untargeted moments

Moment	Data	Model	Source
Block 1: Homelessness			
Share of homeless by age, 25–34 to 65+[a]	26.0, 27.2, 21.0, 18.4, 7.5	26.4, 37.9, 24.2, 10.7, 0.8	HUD PiT 2024
Employed share of the homeless	46	21.4	Meyer et al. (2024)
Income of the homeless, over average	19.1	6.5	Meyer et al. (2024)
Block 2: Who uses and who has a SUD			
Share of users by age, 26–34 to 65+	45, 19, 14, 12, 10	67, 16, 10, 5, 2	NSDUH 2023
Mean age of users, years	39.6	34.6	NSDUH 2023
Unemployment given a SUD	16.2	14.5	NSDUH 2024
Drug use with SMI over without, ratio	2.44	2.64	NSDUH 2024
Drug deaths as a share of deaths, 25–44	20.0	14.5	CDC WONDER
Block 3: Housing and vouchers			
Renters paying over 30% of income	43.2	58.7	SCF 2022
Rent over income, voucher holders	31.2	39.0	HUD PSH 2023
Voucher holders living on wages	28.0	39.2	HUD PSH 2023
Block 4: Income and wealth distributions, shares of the total			
Income, bottom 1, 5, 10, 20%	0.01, 0.31, 0.92, 2.71	0.01, 0.19, 1.00, 3.40	SCF 2022
Net worth, bottom 1, 5, 10, 20, 50%	−0.16, −0.28, −0.29, −0.23, 2.2	0.00, 0.00, 0.00, 0.27, 6.3	SCF 2022
Block 5: Drug price elasticities			
Participation (extensive margin)	−0.3 to −0.9	−1.32	Saffer and Chaloupka (1999), Grossman and Chaloupka (1998)
Intensive margin (users)	−0.3 to −0.8	−0.63	Grossman and Chaloupka (1998), Olmstead et al. (2015)
Block 6: Transmission			
Child of a SUD parent acquires a SUD, odds ratio	1.44, 2.51	1.46	Mellentin et al. (2016), Yoon et al. (2013)
Homeless adult had a homeless parent, odds ratio	1.8 to 2.4	6.23	Bassuk et al. (1997)
Children living with a parent with a SUD, %	4.6	5.8	NSDUH 2024
Block 7: Losing a voucher			
Three-year mortality, households with SMI put on the street, pp[b]	4.6	3.6	Costantini (2025)
Block 8: Deterrence			
Users without a SUD who stop when tested[c]	61	59	Mehay and Pacula (1999)

Notes: percent, except the elasticities and the odds ratios of the transmission block, and percentage points in Block 7; the model is the Housing First benchmark. [a]The share aged 25 to 34 is the target of T_0 in Table 3, so of this row only the other four bands are untargeted. [b]Data: Costantini's instrumental-variables estimate of how much enrolling in the Housing First programme of the Department of Veterans Affairs lowers the three-year mortality of homeless veterans with mental illness. Model: the three-year mortality gap between the households with SMI whom Treatment First's revocation forces into homelessness and the same households, also with SMI, who kept their voucher, with the drug price and π_V held. [c]Data: how much the testing cut past-year marijuana use, against what it would otherwise have been, among service members who stayed in the US military after it began testing at random and dismissing those who failed. Model: of the households that used under Housing First, have no SUD and are subject to the screen (every voucher holder, and every eligible household offered a voucher), the share that stops when Treatment First is introduced, with the distribution, the drug price and π_V held at Housing First's.

similar size is therefore suggestive that, based on the best available evidence, the model does not overstate Treatment First's effect on use.

6 Experiments and results

Every experiment replaces Housing First, the benchmark, with Treatment First, the rule of Section 4.9, and keeps the programme rationed and the tax system as it is in the benchmark. Two objects can respond to the reform: the drug price, by clearing the drug market, and the probability of drawing a voucher π_V, by holding net government revenue at its benchmark level $\bar{R}$, so that what the screen saves is spent on vouchers paid to others. The experiments differ in which of the two respond and in whether the externalities on children are present. Table 5 gives each steady state its own column. There are three experiments, and each answers its own question. The first two experiments compare steady states (Section 6.2).

The first experiment asks whether moral hazard alone justifies conditionality. In economics, moral hazard is defined as the effect insurance has on the incentive to avoid the loss it covers: when part of the cost of a loss falls on someone else, the insured does less to avoid it, which makes the loss more likely and raises what it costs whoever shares in it (Arrow, 1963; Pauly, 1968). The connection to Housing First is that it insures a household against losing its home when drug use leads to a SUD and the SUD to unemployment, and since the assistance does not depend on use, part of what using costs falls on the government's budget, and through it on others who fund the programme. This is the case critics of Housing First make.

To test whether moral hazard justifies Treatment First, I remove the children from welfare and from the parent's preferences, and make their productivity independent of what the parent did. A household's drug use then harms no one but itself and the budget, so any gain comes from correcting moral hazard rather than from correcting the externalities on the children. I then recalibrate the model with Housing First and introduce Treatment First in two different ways, both holding Housing First's drug price fixed. First, π_V

SUD as well, who cannot stop on impact, only 19 percent stop in the model, about a third of the military's response.

is also held at its Housing First level, so that only the screen acts, which shows what deterring users does to the users themselves. Second, with π_V clearing the budget as well, the savings are spent on vouchers for others, which is the gain from correcting moral hazard's cost to the budget that critics of Housing First have in mind.[28] If Treatment First lowers welfare in these cases, moral hazard alone is no reason to make the assistance conditional.

The second experiment introduces Treatment First in the full calibrated model, with the externalities on children, in three steps. Each step lets one more object respond, so that each isolates one thing. The first two repeat the first experiment's two runs, so that each has a counterpart without the externalities. First, the drug price and π_V are both held at their Housing First levels. Therefore, only the screen acts, as in the first experiment's first run, and the one difference between the two runs is whether drug use harms children, so comparing them isolates the externalities: what deterring users does to their children's consumption, housing and adult earnings, net of the children it sends to a shelter. Second, π_V clears the budget, as in the first experiment's second run, so that the screen's savings are spent on vouchers for others. This isolates what that spending is worth once the children are counted. Third, the drug price clears the drug market as well. The drug price comes last because its response is a pure general-equilibrium price effect: the screen lowers the demand for the drug, and the cheaper drug draws users back. This step therefore isolates the general-equilibrium effect of the drug price, and with both objects responding, as they would after a real reform, it becomes the main experiment: whether Treatment First raises welfare as the reform would actually play out.

The third experiment is the transition from Housing First to Treatment First in the full model, with both the drug price and π_V clearing the drug market and the budget, respectively, every year (Section 6.3). The steady states are decades apart, and the households that have a SUD when the reform lands cannot respond quickly, since recovery is slow. The transition establishes how large the early costs of the reform are, who bears them, and whether children born at the reform gain, which is what analysts would see before the long-run effects arrive. Section 6.1 defines the welfare criterion all three use.

The screen. I set the detection rate to $p_{\text{det}} = 85$ percent; office and laboratory urine screens detect 92.5 to 100 percent of opioid use (Manchikanti et al., 2011; Walsh et al., 2025). I set the rate below these, to be conservative, since a lower rate weakens the case for Treatment First.[29] A household that neither uses nor has the disorder is never flagged, since a positive screen is confirmed in a laboratory before it has consequences.

The release rate ψ_x is set to 1/3, so that the expected exclusion is three years. That is the same period for which the Housing Choice Voucher programme already bars a household evicted from federally assisted housing for drug-related criminal activity (Code of Federal Regulations, Title 24, Section 982.553). The test cost c_{test} is set to 0.031 percent of average earnings in the economy, the equivalent of $35 a test. That is the upper end of the estimates: the Texas Legislative Budget Board's fiscal note for Senate Bill 11 puts a test at $35 (Legislative Budget Board, 2013), which again loads the experiment against Treatment First.

6.1 The welfare criterion

Every welfare number in this paper is a consumption-equivalent variation (CEV), computed as follows. Write V^b for expected lifetime value under the benchmark and V^c for expected lifetime value under a counterfactual policy. Both are taken at birth, before any of the newborn's draws are realised, including the household it is born into, and each under its policy's own distribution of newborns, which the policy moves, since a child's starting productivity depends on what happened to its parent.[30] The CEV Δ is the proportional change in non-housing and non-drug consumption, applied at every date and in every state, that makes a newborn

[28] I do not let the drug price clear in this experiment, because its response is not part of moral hazard: it is set in the drug market, and a fall in it is a gain to every user, assisted or not, that has nothing to do with the incentives the assistance creates. Letting it clear would also turn the question into one about the elasticity of the drug supply, which I take to be low for the reasons given in Section 5.1, while other studies take it to be high, treating the supply of an illegal drug as competitive at a constant unit cost (Becker et al., 2006). Section 6.2 reports what letting it clear would add.

[29] The qualitative results continue to hold with a detection rate of 70 percent and with a more elastic supply of the drug.

[30] The ex-ante expected newborn welfare is the standard criterion in this literature (Conesa et al., 2009; Guvenen et al., 2023).

under the benchmark as well off as the same newborn under the counterfactual:

$$V^b\big((1+\Delta)\,c\big) \;=\; V^c(c). \tag{30}$$

A positive Δ therefore means the counterfactual is worth that much of a permanent consumption gain, and a negative one that it costs that much.

Whose welfare. Welfare here is measured for newborns, and it includes childhood in the parent's household and adult life after it. For a child born in period t it is

$$\begin{aligned} W_t \;=\; & \frac{1}{\Phi_t(1,\mathcal{S})}\sum_{j=1}^{J_K}\beta^{\,j-1}\int_{s^{\text{parent}}\in\mathcal{S}}\Big\{(\xi(j)-1)\,\big[\,U\big(X_c(j,s^{\text{parent}})\big)+\aleph\,\big] \\ & +\;\beta\,\chi(j)\int_{s^{\text{child}}\in\mathcal{S}}V_{t+j}(1,s^{\text{child}})\,\mathcal{N}\big(\mathrm{d}s^{\text{child}}\mid s^{\text{parent}}\big)\Big\}\,\mathrm{d}\Phi_{t+j-1}(j,s^{\text{parent}}), \end{aligned} \tag{31}$$

where $V_t(j,s)$ and $\Phi_t(j,s)$ are the value and the distribution of Section 4.10 and Section 4.13 in period t, the subscript being the date and j the age. The outer integral is over the state s^{parent}, including the disorder d, of the parents of age j in period $t+j-1$, and the inner one over the state s^{child} the child enters with, given its parent's, so that together they run over every pair of parent and child states. $\Phi_t(1,\mathcal{S})$ is the mass of households that entered in period t.

The first term is childhood. A household joins the workforce at $j=1$ with its child just born, so the households of age j in period $t+j-1$ are the surviving parents of the children born in period t, each child in its j-th year at home, and $X_c(j,s^{\text{parent}})$ is the bundle the child shares with its parent who is in state s^{parent}. The child's own flow utility from that bundle, with the flow value of being alive, is counted as $\xi(j)-1$ of the household, as it is in the parent's problem. A child lives at home only while its parent is alive, so each year of its childhood is weighted by the share $\Phi_{t+j-1}(j,\mathcal{S})/\Phi_t(1,\mathcal{S})$ of parents still alive, and discounted to its birth.

The last term is adult life, which begins when the child leaves home: at the end of period $t+j-1$, when its parent is of age j, with the probability $\chi(j)$ of the parent's problem (24), the parent's chance of dying in that year before J_K, and one at J_K. The child then joins the workforce at model age $j=1$ in period $t+j$, placed by $\mathcal{N}$ as in (29), with the value $V_{t+j}(1,s^{\text{child}})$, discounted by the years it spent at home.

The child's own childhood counts at weight one here, not at its parent's weight ω_c, and that is what brings the externality on the child into welfare. Its adult part is the value its own problem maximises, so the child it raises in turn enters there at the weight ω_c. In a steady state the dates drop out.

Two horizons are used, and they are the same object measured at different dates: a newborn on the date the policy is implemented, who lives through the transition, and a newborn in the new steady state.

6.2 The long run

Table 5 sets the long run under Treatment First beside the benchmark, and, in the first three columns, Housing First and Treatment First without the externalities.

Treatment First without externalities. To capture the no-externality case, I treat the child as appearing only as an adult: its parent does not weigh it, its productivity does not depend on what its parent did, and its childhood, the years its parent is 26 to 50, is not counted in welfare. I recalibrate the model to this configuration.

With the drug price and π_V held at their levels under Housing First, Treatment First cuts drug use by 60 percent and the share with a SUD by 61 percent, against Housing First with the externalities switched off, while homelessness rises by 3 percent (first two columns of Table 5). And yet, no household is better off under Treatment First, with newborns losing 1.50 percent of lifetime consumption as adults. These losses capture the value of the option removed from a rational, forward-looking agent.

Rebalancing the budget does not change the ranking. When π_V instead rebalances the budget at Housing First's drug price (third column), the screen's savings buy more vouchers. Therefore, π_V rises from 6.75 to

6.89 percent, and 63 percent of households gain, but by 0.14 percent of consumption on average. The gain is therefore small and spread thin, while the loss is concentrated on the users the screen takes out: 96 percent of households with a SUD are worse off, and newborns still lose 1.31 percent of lifetime consumption as adults (since childhood is not counted here). The impact of moral hazard on the budget is still not enough to outweigh the loss of that option to households that choose optimally.[31]

Treatment First with externalities. Having ruled out moral hazard as a reason for sobriety conditions, I now test, in the three steps set out above, whether the externalities would be a reason to impose them. With the drug price and π_V both held at their Housing First levels (fifth column of Table 5), only the screen acts, as in the second column: drug use falls by 64 percent, homelessness is unchanged, and the share of children in a shelter rises from 1.03 to 1.04 percent. Yet a newborn now gains 5.04 percent, where without the externalities it loses 1.50. The externalities alone turn the loss into a gain: the cut in drug use reduces the *budget and earnings externalities*, as the children of the users the screen deters get more of their parents' consumption and housing and start adult life more productive, and that outweighs the *childhood homelessness externality* of the few more children it sends to a shelter.

Letting π_V clear the budget at Housing First's drug price (sixth column) spends the revenue the screen frees on vouchers: π_V rises to 11.45 percent, drug use falls by 74 percent, homelessness falls by 23 percent, the share of children in a shelter falls by 21 percent, and a newborn gains 13.11 percent. Drug use falls by more than with both held because the higher π_V makes receiving a voucher more likely, and thus staying clean worth more. Childhood homelessness falls because the additional spending on vouchers lifts other low-income households of child-rearing age, which do not use drugs, out of homelessness.

Finally, letting the drug price clear the drug market as well gives the main experiment (last column). Under this configuration, Treatment First cuts drug use by 32 percent, from 8.96 to 6.09 percent of households, and the share of households with a SUD by 33 percent, from 3.81 to 2.54 percent of households. It also raises the share of households with a SUD who are in treatment, from 24.9 to 29.3 percent, because treatment buys a higher chance of recovering, and Treatment First makes recovering worth more: a household that has recovered and does not use passes the screen and can hold a voucher if it is eligible and offered one. As users who lose their vouchers leave the programme, it costs less at a given π_V. Tax revenue also rises, mainly because more children start adult life with higher productivity, and to a lesser extent because fewer adults lose their jobs to a SUD. Thus, π_V rises from 6.75 to 9.62 percent, by 42 percent, making staying clean of drugs even more desirable. At the same time, taking users out of the drug market lowers the drug price by 14 percent, which draws some households back.

Treatment First also lowers homelessness by 1 percent, from 0.62 to 0.61 percent of households, the result of the balance of three effects. First, it revokes the vouchers of households with a SUD and puts them at a higher risk of homelessness. Of them, 5.63 percent are homeless, against 3.28 under Housing First. On its own this raises homelessness by 10 percent. Second, less SUD means less unemployment and so less homelessness. There are a third fewer households with a SUD, which on its own lowers homelessness by 6 percent. Third, through the higher π_V, more vouchers go to other clean and eligible households, lowering homelessness by 5 percent. Together the last two more than offset the first. For households aged 26 to 50, the years a child lives with them, the same three effects nearly net to zero, so the share of children in a shelter remains unchanged at 1.03 percent. These are long-run outcomes, however. In the first years after the reform both homelessness and childhood homelessness rise, because the households that already have a SUD lose their vouchers within a year, while their number falls only as fast as recovery allows. I take the transition up in Section 6.3.

Turning to welfare, a newborn is better off under Treatment First by 4.33 percent of lifetime consumption. This step isolates the general-equilibrium effect of the drug price: against the second step, where the drug price was held at its Housing First level, the cheaper drug draws users back, and since their use harms their children, the gain falls from 13.11 to 4.33 percent.[32] About 57 percent of the gain in lifetime utility comes in

[31] If the drug price is also allowed to clear the drug market, newborns gain 0.39 percent as adults. That is a general-equilibrium effect of the drug price and has nothing to do with moral hazard: the screen lowers demand, the cheaper drug draws users back, and more of them are caught and lose their vouchers, which the budget then gives to others.

[32] Letting the drug price clear with π_V held instead lowers it by 13 percent; the revenue the screen frees goes unspent, drug use falls by 27 percent, homelessness rises by 13 percent, and a newborn gains 0.18 percent.

Table 5: The long run under each policy

	No externalities			With externalities			
		Treatment First			Treatment First		
	Housing First	both held	π_V clears	Housing First	both held	π_V clears	both clear
Adults							
Drug price, benchmark = 1	1.03	1.03	1.03	1.00	1.00	1.00	0.86
Voucher probability π_V	6.75	6.75	6.89	6.75	6.75	11.45	9.62
Drug use	9.01	3.59	3.58	8.96	3.20	2.31	6.09
Carrying a SUD	3.84	1.51	1.51	3.81	1.35	0.98	2.54
Homelessness	0.60	0.62	0.62	0.62	0.62	0.48	0.61
SUD share of the homeless	20.5	16.1	16.2	20.2	14.7	13.7	23.3
In treatment, of those with a SUD	25.0	29.9	29.9	24.9	29.9	30.0	29.3
Children							
Living with a parent who uses drugs	16.42	6.57	6.57	16.32	5.87	4.25	10.99
Living with a parent who has a SUD	5.84	2.32	2.32	5.81	2.08	1.52	3.89
In a shelter	1.01	1.04	1.03	1.03	1.04	0.82	1.03
Welfare							
CEV of a newborn, percent of lifetime consumption	–	−1.50	−1.31	–	5.04	13.11	4.33
Share of the newborn's gain that comes in childhood	–	–	–	–	89	74	57
Households entering adult life at 26 better off	–	0	3	–	0	100	100
All households better off	–	0	63	–	0	74	76
Households with a SUD better off	–	0	0	–	0	0	20
Newborns better off, by the household they are born into	–	–	–	–	83	99	92

Notes: percent unless the row says otherwise; each column is a steady state. The first three columns are Housing First and Treatment First without the externalities, with the child appearing only as an adult. Under Housing First the drug price clears and π_V is the benchmark's. Under Treatment First, the drug price and π_V are both held there in the second column; and the drug price is held and π_V clears the budget against the first column's net revenue in the third. The CEV in the second and third columns is the newborn's value at 26, against the first column. Housing First, the fourth column, is the benchmark with the externalities. Treatment First screens with a detection rate of 85 percent. In the fifth column the drug price and π_V are both held at their Housing First levels, in the sixth the drug price is held and π_V clears, and in the last both clear; when π_V is held, the revenue the screen frees goes unspent, and when it clears, π_V holds net government revenue at its benchmark level. Children are those living with households aged 26 to 50, the years a child lives at home. The CEV is against the benchmark. A household is better off when its value under Treatment First exceeds its value in the same state under Housing First, using the Housing First distribution.

childhood. The share of children living with a parent who uses drugs falls from 16.32 to 10.99 percent, and the share living with one who has a SUD falls from 5.81 to 3.89 percent. As a result, parents spend more on the consumption and housing they share with the child and less on the drug, and they are less likely to acquire a SUD and remain unemployed because of it, raising childhood consumption further.

The other 43 percent comes in adult life, from two sources. The first, 11 percent of the whole gain, is the better productivity a child draws when its own parent does not carry the disorder. Escaping the penalty is worth a lot to each child it spares, about 12 times what the average child gains at the same productivity. But it spares only about 3 percent of children, since the share of newborns who enter adult life with the penalty falls from 8.99 to 6.07 percent.

The second, the remaining 32 percent, does not come from the child's productivity at all. It is a result of the better economy it lives its adult life in. A clean and eligible household is 42 percent more likely to draw a voucher. And a voucher is worth most to a household that enters adult life with low productivity, because it is the likeliest to need one, as the value of entering adult life rises about 30 times as much at the lowest productivity as at the highest. This part of the gain is therefore larger for a child whose own parent carried the disorder, since the penalty leaves it with lower productivity. Moreover, even a child who goes on to use drugs pays less for them: the drug is 14 percent cheaper in equilibrium. Thus, every household that enters adult life at 26 is better off under Treatment First than it would be in the same state under Housing First (Table 5). Of newborns (including childhood), 92 percent are better off under Treatment First than they would be if born into the same household under Housing First. Across all households, 76 percent are better off and 3 percent worse off; the rest, all older than 50, are essentially unaffected. Finally, even 20 percent of those that have a SUD are better off, due to the lower drug price.

6.3 The transition

To see what the reform does on the way to the long run, I compute the transition from Housing First to Treatment First. The transition starts from the benchmark, with the reform landing as an unexpected permanent shock at the start of the first year, which I simulate for seventy years all the way to the Treatment First steady state. The drug market and budget clear every year. Figure 2 shows what the transition looks like along eight key dimensions.

Panels (a) and (d) show drug use and the share of parents who use, respectively. Drug use falls upon impact: 84 percent of its long-run fall happens in the first year. The share of parents using drugs falls from 16.32 to 11.31 percent, above its long-run level of 10.99 percent. These shares drop because the screen acts on the decision to use from the day the reform lands, and households who do not have a SUD have agency over whether they use drugs or not. The screen affects households both directly, since a voucher holder that uses risks losing its voucher within a year, and indirectly through expectations, since a household that uses risks having a voucher offer revoked later, when it may need one but have a SUD, while a disorder takes years to clear.

Panel (b) shows the share with a SUD. This share falls slowly, because recovery is slow: half of its long-run fall is done by year 11, and nine tenths by year 27. That stock keeps demand for the drug up, and the drug price stays above its long-run level for 36 years (panel (c)). The elevated price deters households without a SUD, at every age: in the first decade the number of drug users without a SUD falls by up to 47 percent, against 31 percent in the long run. The share of parents who use stays above its long-run level all the same, because it counts the parents who have a SUD, whose number falls only slowly (panel (b)).

Panels (e) and (f) show homelessness and the share of children in a shelter, respectively. Homelessness overshoots. It rises by 15 percent at its peak, in year 7, and then declines slowly towards its long-run level, 1 percent below that under Housing First. A similar pattern is observed for childhood homelessness: the share of children in a shelter rises from 1.03 to 1.16 percent in year 7, against 1.03 percent in the long run. It rises for the same reason: the parents who have a SUD when the reform lands lose their vouchers within a year, and it comes back down only as their number falls.

Panel (g) shows the CEV of each cohort by its year of birth. Despite the increase in childhood homelessness, children born at the reform still gain about 2.7 percent of lifetime consumption, although less than those born in the long run. They gain, because the fall in parental drug use reaches far more children than the overshoot in homelessness and childhood homelessness: in the first year the share of parents who use drugs falls by 5.01 percentage points, while the share of children in a shelter rises by at most 0.13 points, too few children to reverse their gain. They gain less than in the long run, because they grow up while the share of parents with a SUD is still close to its Housing First level, since that share falls only as fast as recovery allows, so more of them spend their childhood with a parent who has the disorder and in a shelter, and more enter adult life with the earnings penalty. The gain stays near that level for children born while that share is still falling, and rises toward the long run for those born later, whose parents were themselves born years after the date the reform was implemented.

Of the adults alive at the reform, only 3 percent are worse off under it, and even 14 percent of those who have a SUD when it lands benefit, because the lower drug price more than offsets the loss from the sobriety requirement for some of them. Panel (h) repeats the comparison in every year of the path, for a household in each state weighted as under Housing First. The share worse off stays at about 3 percent, and the share of those with a SUD who are worse off falls from 86 to 80 percent as the price comes down further. Every household entering adult life at 26 is better off throughout. A newborn's comparison also counts its childhood, which may be spent with a parent who loses a voucher, so fewer newborns than entering adults gain: in the long run, 92 percent are better off than if born into the same household under Housing First (Table 5).

7 Discussion and implications

To close, I return to the question I started with: should housing assistance require sobriety? The answer turns largely on the children. The moral hazard that dominates the debate is real in the model, which I calibrate

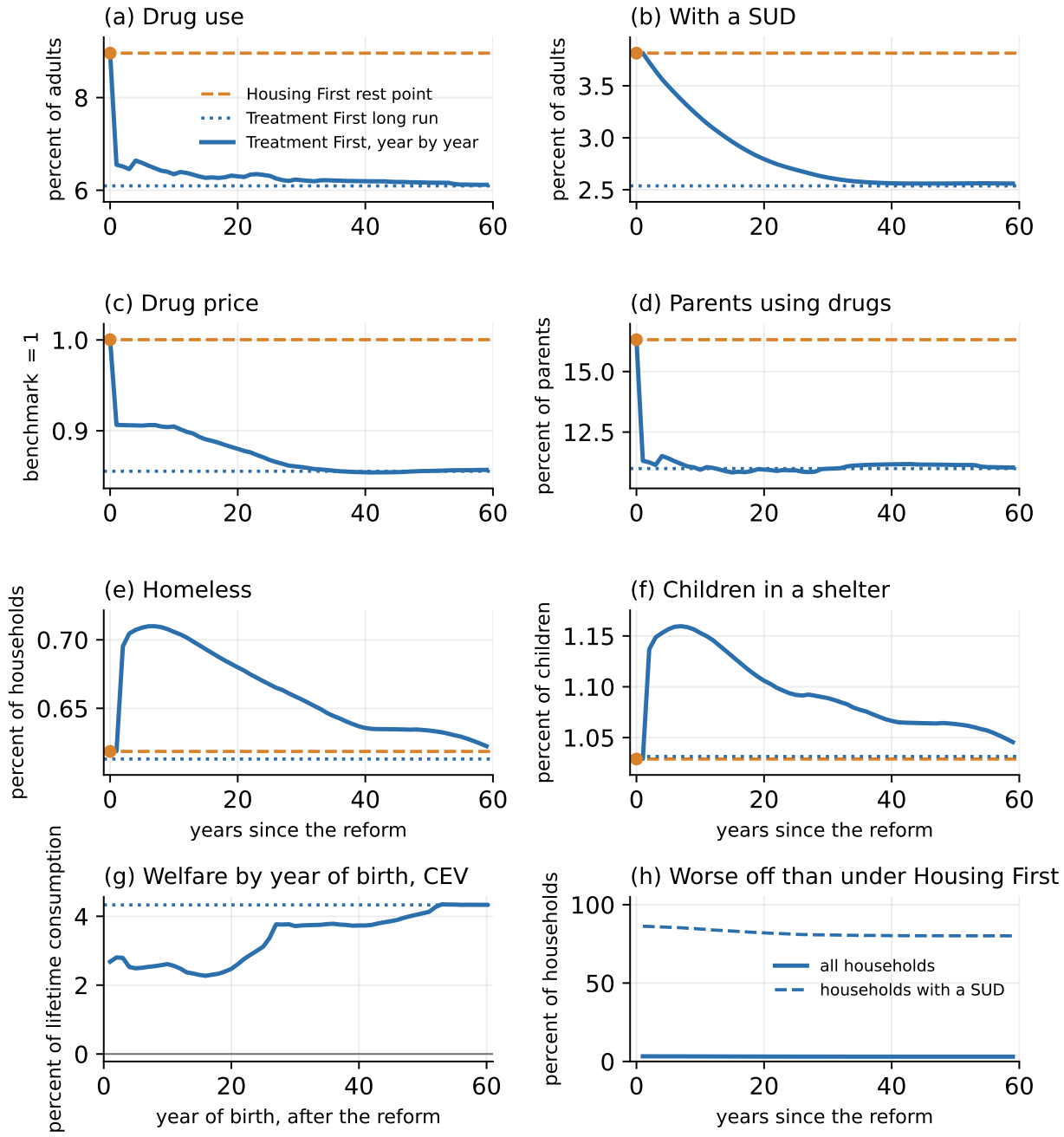


Figure 2: The transition from Housing First to Treatment First, quantitative model

Notes: panels (a) to (f) and (h) are by year since the reform, which lands at the start of year 1, with the benchmark as year 0; the dashed orange line is the Housing First benchmark, the dotted blue line the Treatment First long run and the solid blue line the path. Children are those of households aged 26 to 50. Panel (g) is the CEV of each cohort born on the path, against the benchmark, by its year of birth. Panel (h) is the share of households worse off in each year of the path than in the same state under Housing First, weighted by the benchmark's distribution.

to US data: without the externalities on children, drug use under Treatment First falls by 60 percent, close to the 61 percent fall that random testing brought about in the US military (Mehay and Pacula, 1999). Yet a newborn is worse off by 1.31 percent, even when the screen's savings fund more vouchers. This is because deterring households that weigh the risks correctly removes an option they value, while the impact on the budget is not large enough. Thus, correcting moral hazard is no reason to make housing assistance conditional.

What drives the kinds of gains policymakers would actually be concerned about is the externalities drug use imposes on children: the *budget and earnings externalities* that Housing First leaves in place outweigh the *childhood homelessness externality* that Treatment First creates. In the run in which only the screen acts, with the drug price and π_V held, a newborn gains 5.04 percent with the externalities and loses 1.50 percent without them, while the share of children in a shelter rises only from 1.03 to 1.04 percent.

In the long run with externalities, and with the drug price and π_V both clearing, Treatment First cuts drug use by 32 percent and the share of households with a SUD by 33 percent, and actually lowers homelessness by 1 percent. As a result, it raises the welfare of a newborn by 4.33 percent of lifetime consumption (childhood included). Most of that gain comes in childhood, from parents who spend less on drugs and are less likely to remain unemployed because of a SUD after losing their job. The rest comes from higher earnings and from a larger voucher programme, which the users it excludes and the higher revenues from the taxes paid by more productive and more employed households pay for.

Children born at the reform gain less than those born in the long run, because they grow up while the share of parents with a SUD is still close to its Housing First level, and are thus also more exposed to the *childhood homelessness externality*, but they still gain. Yet, from a policy point of view the transition can be deceptive. The households that have a SUD when the reform lands lose their vouchers within a year, so homelessness first rises, by up to 15 percent, and the share of children in a shelter with it. It comes down only as the number with a SUD falls, which takes 27 years to complete nine tenths of its fall.

Three lessons follow. First, the arguments for Housing First and Treatment First made by different advocacy groups leave out the outcome that matters most: what happens to the children of the households a sobriety requirement deters from using, in what they consume while at home and in what they earn as adults. If this framing continues to dominate the discussion, the reform will most likely be judged on the wrong outcomes.

Second, the politics of the reform can run against its own payoff. Its apparent costs would come first, falling on households already having a SUD, and would appear in the homelessness and childhood homelessness counts during the transition from the Housing First benchmark to the long-run equilibrium under Treatment First; its positive effects on homelessness would only come later. Judged even on its first 25 years, by the PiT count, the reform will be most exposed to reversal before its long-run positive effects on homelessness, the SUD and, most importantly, children have arrived in full.

Third, the results point to two choices of design that the current debate leaves out. The first is what to do with the excess funds the government collects under Treatment First. In my experiment, all of them go back into vouchers, so an eligible household that does not use drugs is 42 percent more likely to get one, which is one of the reasons a newborn's adult life is worth more, and homelessness actually declines slightly in the long run, with childhood homelessness unaffected. A sobriety requirement introduced without recycling that excess revenue back into more vouchers would give up nearly all of the gain: with π_V held at its benchmark level, a newborn would gain 0.18 percent rather than 4.33, and homelessness would rise by 13 percent.

The second is whom the condition applies to, which may further mitigate the impact on the households that already have a SUD along the transition. The gain comes from deterrence: adults who do not have a SUD and are able to stay clean do so to keep their access to a voucher. The early cost falls on a different group, the households that already have a SUD when the reform lands. They cannot respond quickly, because recovery from a SUD is slow, so for them the condition brings mostly the cost. Their homelessness rises within a year, and 86 percent of them are worse off. A condition applied only to new voucher holders, with current holders given a five-to-ten-year grandfather exemption, for example, may strike a good balance between keeping the threat for those who can still respond to it and sparing those who cannot.

A Calibration of the standard parameters

The remaining thirty-three parameters are more conventional, and I record the values I use so that the calibration can be reproduced. Six of them I calibrate internally, against a moment, because no source reads them off directly; the other twenty-seven I set from outside, from an estimate or from statute. Table A1 and the entries below run in the same seven blocks, and each entry gives the moment or the source that fixes the parameter together with the value that moment takes in the data. The Model column reports what the model produces against that value, and it does so only for the six I calibrate internally, since the other twenty-seven have no target to reproduce.

The parameters in this appendix are not inert in the experiments. For example, the discount factor governs how heavily a household weighs a death later in life, the dispersion of productivity determines how many households are poor enough to be at risk of homelessness and of acquiring a disorder, and risk aversion governs what insurance is worth. What separates them from the parameters above is not that they do nothing, but that they are not specific to this paper.

Table A1: Standard parameters

Parameter	Description	Set	Moment and source	Data	Model	Value
Block 7: Preferences						
σ	Risk aversion	Ext.	Attanasio and Weber (2010)	2.00	n.a.	2.00
γ	Weight on housing services	Int.	Housing over expenditure net of drugs, CEX 2024	26.8%	26.2%	0.536
β	Discount factor	Int.	Wealth over output, BEA fixed assets	3.00	2.95	0.946
$\xi(j)$	Equivalence scale	Ext.	Square-root scale, OECD	1.41	n.a.	1.41
Block 8: Rental ladder						
h_2	Smallest rental size	Int.	Renters paying over half of income, SCF 2022	19.2%	25.8%	0.0407
h_3/h_2	Middle rung over the smallest	Ext.	Set	4.00	n.a.	4.00
h_4/h_2	Largest rung over the smallest	Ext.	Set	8.00	n.a.	8.00
Block 9: Earnings and employment						
J	Maximum age	Ext.	Age 85	60	n.a.	60
J_K	Years the child lives at home	Ext.	Parent aged 26 to 50	25	n.a.	25
J_R	Retirement age	Ext.	Age 66, SSA full retirement age	41	n.a.	41
$\varepsilon(j)$	Age profile of efficiency	Ext.	Hump peaking at age 56, Greenwood et al. (2025)	Profile	n.a.	Profile
ρ_y	Persistence of productivity	Ext.	Guvenen et al. (2023)	0.90	n.a.	0.90
σ_ζ	Standard deviation of its shocks	Int.	Bottom half's share of earnings, SCF 2022	17.2%	16.2%	0.413
ρ	Persistence of productivity across generations	Ext.	Solon (1992), Mazumder (2005)	0.50	n.a.	0.50
$\bar{\pi}^{eu}$	Job separation rate	Int.	Unemployment rate, BLS CPS 2014–24	4.9%	5.6%	0.0268
$\varpi(y)$	Separations by productivity	Ext.	Unemployment by education, BLS 2019	Profile	n.a.	Profile
$\bar{\pi}^{ue}$	Annual job finding rate	Ext.	Krueger et al. (2014)	0.55	n.a.	0.55
p_{ui}	Share of separations that receive UI	Ext.	UI applicants among the unemployed, BLS 2022	0.30	n.a.	0.30
Block 10: Baseline survival and SMI onset						
$\bar{\phi}(j)$	Baseline survival by age	Ext.	US life table 2010, Arias (2014)	Profile	n.a.	Profile
$\varsigma_m(j)$	Age profile of SMI onset	Ext.	SMI prevalence by age, NSDUH 2024	Profile	n.a.	Profile
Block 11: Interest and taxes						
r	Risk-free rate	Ext.	Kaplan et al. (2020)	0.03	n.a.	0.03
φ	Tax progressivity	Ext.	Heathcote et al. (2017)	0.85	n.a.	0.85
τ_ℓ	Level of the labour income tax	Int.	Average labour income tax, McDaniel (2007)	22.4%	20.9%	0.894
τ_k	Capital income tax rate	Ext.	US long-run rate on gains and dividends	0.15	n.a.	0.15
Block 12: Social security benefits and programmes' rules						
$S(y)$	Social security	Ext.	SSA primary insurance amount	Statute	n.a.	0.90, 0.32, 0.15
$\underline{S}$	SSI floor over average labour income	Ext.	SSA federal benefit rate	0.177	n.a.	0.177
ψ_{ui}	UI replacement rate	Ext.	Department of Labor, statutory rates	0.50	n.a.	0.50
$\bar{y}$	Cap on the UI earnings base	Ext.	Department of Labor, state benefit maxima	1.00	n.a.	1.00
κ_V	Voucher means test	Ext.	HUD rule	0.50	n.a.	0.50
κ_r	Tenant's rent, share of income	Ext.	HUD rule	0.30	n.a.	0.30
ϱ	Food stamp clawback rate	Ext.	SNAP statute, 7 CFR 273.10	0.30	n.a.	0.30
Block 13: Drug market						
J_E	Maximum age at which use can begin	Ext.	Model age 55, that is age 80	55	n.a.	55
ν_1	Supply shifter	Ext.	Normalisation, so $p_v = 1$	Norm.	n.a.	Norm.

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

Block 7: Preferences. The four parameters in this block determine what a household wants, and what it is willing to give up to get it.

σ, risk aversion. It sets the marginal utility of consumption, so it affects every welfare comparison in the paper. Increase it and households want smoother consumption and insurance is worth more, which

then means that the voucher is worth more. I set it to the standard value of two, following Attanasio and Weber (2010).

γ**, the weight on housing services.** It is the weight on housing in the bundle of consumption and housing, so it sets what a dwelling is worth against consumption, and that bundle is what the drug is weighed against. As γ rises housing is worth more relative to consumption, households buy more of it, and the share of spending that goes on housing rises with it. γ is chosen for the share it produces: shelter plus utilities take 26.8 percent of expenditure in the 2024 CEX, which excludes illicit drug spending.

β**, the discount factor.** It sets how much households discount the future, and thus how much they save and how much the prospect of death deters them from using drugs. As it rises households save more, the wealth-to-output ratio rises and households are more deterred by death. It is chosen so that wealth over output matches the national accounts, a ratio of three, from the Bureau of Economic Analysis (BEA) fixed asset tables against National Income and Product Accounts (NIPA) output, both read at the 2023 vintage, the last year the fixed asset tables cover.

$\xi(j)$**, the equivalence scale.** The household's consumption and dwelling are shared with the child while it lives with its parent, and thus are divided by $\xi(j)$ to give what each member gets. Raise it and each member gets less of a given budget, and the child counts for more of the household, through the weight $\xi(j) - 1$ in (12). It is the square-root scale (OECD, 2013), $\sqrt{N(j)}$, where $N(j)$ in the model is set to two while the child lives with the parent ($j \leq J_K$), from the parent's age 26 to 50, and to one after the child has left ($j > J_K$).

Block 8: Rental ladder. These three parameters build the grid of dwelling sizes a household chooses from. Below the smallest rung is the shelter, h_1, which was calibrated in Block 2.

h_2**, the smallest rental size.** The size of the cheapest dwelling is what decides how much of its income a poor household must spend to be housed at all. Raise it and the bottom rung is larger and dearer, so more households are burdened, and a voucher is worth more to them. It is chosen so that the fraction of renters paying more than half of their income for housing is similar to the data, 19.2 percent in the 2022 SCF; the model gives 25.8 percent.

h_3/h_2 **and** h_4/h_2**, the spacing of the ladder.** The second and upper rungs set how much housing they provide, and so what a household that can afford more than the smallest unit can move up to. Raise them and each step up the ladder costs more rent, and fewer households can rent them. The two larger rungs are set at 4.00 and 8.00 times the smallest.

Block 9: Earnings and employment. These ten parameters determine how long a household lives and works, what it is expected to earn over its life, how persistent the shocks to it are, and how it moves between work and unemployment.

J**,** J_K **and** J_R**, the length of life, of child rearing and of working life, respectively.** A household lives at most J periods, has its child at home for its first J_K of them and works until age J_R, after which it retires on social security. Together they set how many years of life and of earnings a disorder can cost, and J_K how long a parent's drug use reaches their child through what they share. Raise either J or J_R and a disorder that shortens life or work costs more; raise J_K and the *budget and childhood homelessness externalities* last longer. I set $J = 60$, so that with $j = 1$ at age 26 the last age is 85, $J_K = 25$, so that the child lives at home while its parent is 26 to 50, and $J_R = 41$, so that retirement begins at age 66, the full retirement age under social security for those born between 1943 and 1954.

$\varepsilon(j)$**, the age profile of efficiency.** This is the deterministic hump in earnings over the life cycle. Without it every household with the same productivity would earn the same at every age, and the ages at which households are poor, and so at risk of homelessness and of using, would be wrong at both ends of working life. I take it as the exponential of a quadratic in age used by Greenwood et al. (2025), $\varepsilon(j) = \exp\{-(j-1)^2/1800 + (j-1)/30\}$ with $j = 1$ at age 26. It peaks at age 56, having risen 65 percent from age 26, and then falls 4 percent from the peak to age 65, the last working age.

ρ_y **and** σ_ζ**, the productivity process.** Log productivity is persistent but not permanent: it follows an autoregressive process with persistence ρ_y and shocks with standard deviation σ_ζ. σ_ζ is what decides

how many households are poor enough to be at risk of homelessness, and the share of earnings that goes to the bottom half of working households falls in it. I set $\rho_y = 0.90$, the persistence of the income process in Guvenen et al. (2023). σ_ζ is calibrated to the bottom half's share of earnings among households aged 26 to 65 with positive wage income in the 2022 SCF, 17.2 percent.

ρ**, the persistence of productivity across generations.** The child's log productivity at entry into adulthood is drawn around its parent's with weight ρ and around the mean with weight $1 - \rho$, so ρ is how much of its parent's current productivity a child keeps. It is the channel through which anything that lowers a parent's productivity lowers their child's: a parent whose own productivity carries the penalty δ_d, because their own parent had a SUD, also passes a lower draw on to their child, so a parent's SUD lowers productivity beyond the next generation. Raise it and a child's prospects depend more on its parent's decisions, and the *earnings externality* grows. It is set externally at 0.50, between the intergenerational elasticities of earnings in Solon (1992) and Mazumder (2005).

$\bar{\pi}^{eu}$**, the job separation rate.** This parameter determines the chance that an employed household loses its job each year. It is scaled by $\varpi(y)$ in (21), so separations are likelier at low productivity. Raise it and more households lose work each year, so the unemployment rate rises with it. It is chosen to reproduce the aggregate unemployment rate in the Current Population Survey (CPS) of the Bureau of Labor Statistics (BLS) over 2014 to 2024, 4.9 percent, given the job-finding rate set externally below.

$\varpi(y)$**, separations by productivity.** This parameter tilts the separation rate across productivity, so that low-productivity households lose their jobs more often, which is what puts them at a higher risk of homelessness. Steepen the tilt and job loss concentrates further among the poorest, and so does the risk of homelessness. It is normalised to average one across households, so that $\bar{\pi}^{eu}$ alone sets the aggregate unemployment rate. Each productivity node takes the unemployment odds of the education group it falls in, from the BLS 2019 annual averages of unemployment by educational attainment for ages 25 and over, with education ordering productivity by median earnings.

$\bar{\pi}^{ue}$**, the annual job-finding rate.** This parameter is the probability that an unemployed household finds work each year it is unemployed. With the divisors κ_d^{ue} and κ_m^{ue} it sets how long an unemployed household with a SUD or SMI is expected to remain unemployed. Raise it and spells out of work are shorter on average, so fewer households fall behind on the rent. I set it to 0.55 from the annual exit rate Krueger et al. (2014) estimate on matched CPS panels.

p_{ui}**, the share of separations that receive UI.** A household that loses its job draws UI for one year with probability p_{ui} and nothing otherwise. It sets how many of the newly unemployed fall straight to zero earnings, and how costly being unemployed is. I set it to 0.30, since about seven in ten unemployed people who worked in the past year did not apply for UI after losing their job in the BLS supplement to the CPS for 2022.

Block 10: Baseline survival and SMI onset. These two parameters are the survival and the SMI-onset age profiles.

$\bar{\phi}(j)$**, baseline survival by age.** This is the baseline chance of surviving to the next age, before any multiplier. It is the profile every mortality multiplier in Section 5 multiplies, so it is the base the SUD, homelessness and SMI hazards scale. Raise it and households expect to live longer, so the life a SUD puts at risk is worth more. I take the profile from the life table for the total population of the United States in 2010 (Arias, 2014), for ages 26 to 85.

$\varsigma_m(j)$**, the age profile of SMI onset.** This determines the shape of onset across ages, normalised to one at age 26, so that ω_m in Table 3 sets its level alone. It is here rather than in Table 3 because, like the age survival profile, it only sets the baseline; what actually affects the impact of Housing First versus Treatment First is ι, the multiplier a SUD puts on SMI onset, which is in Table 3. It is the onset that reproduces a log-linear fit to the share of adults carrying SMI at each age in NSDUH 2024.

Block 11: Interest and taxes.

r**, the risk-free rate.** This parameter sets the return on saving. Raise it and saving pays more, so households hold more assets to carry them through a spell out of work or without a voucher. I set it to three

percent, from Kaplan et al. (2020).

φ**, tax progressivity.** This is the exponent on income in the tax schedule of Section 4.3, and it determines the progressivity of the labour income tax. Lower it and the tax is more progressive, so households with low incomes, among them most voucher holders and most drug users, keep more of what they earn. It is set to 0.85, in line with Heathcote et al. (2017).

τ_ℓ**, the scale of after-tax labour income.** Given φ, this parameter sets how much of their labour income households keep after tax on average, so raising it lowers the tax liability. It does not move in the experiments: the budget is balanced by π_V instead. It is calibrated to an average labour income tax rate of 22.4 percent, which is what McDaniel (2007) measures for the United States and what Guvenen et al. (2023) use.

τ_k**, the capital income tax rate.** This parameter affects the after-tax return on savings. Raise it and saving pays less, so households hold fewer assets to fall back on. I set it to fifteen percent, consistent with the statutory United States rate on long-term capital gains and qualified dividends for the middle of the income distribution, which the Internal Revenue Service (IRS) sets out in Topic 409 (`https://www.irs.gov/taxtopics/tc409`).

Block 12: Social security benefits and programmes' rules.

$S(y)$ **and** $\underline{S}$**, social security and its floor.** Retirement income follows the statutory primary insurance amount: it replaces a share of the household's income at its last working age that falls across the two bend points of the Social Security Administration (SSA), up to the taxable maximum, and with the SSI payment as a floor (the full function is detailed in footnote 11). As $S(y)$ rises the old are richer, so the value of a life still to be lived rises and with it the cost of a disorder that shortens it. The shares $\alpha_1, \alpha_2, \alpha_3$ are $0.90, 0.32, 0.15$ below the first bend point, between them and above the second, respectively. The bend points k_1 and k_2 and the taxable maximum $y^{\max}$ are the SSA's 2024 dollar amounts divided by its average wage index, 0.22, 1.33 and 2.64, respectively. The floor, $\underline{S}$, is 0.18 of average labour income.

ψ_{ui} **and** $\bar{y}$**, UI.** These two set the UI replacement rate and the cap on the earnings it is paid on, respectively. Raise either and the first year out of work costs a household less, so a spell out of work is less likely to end in a shelter. I set them to fifty percent and one times average labour income, respectively, taking both from the statutory replacement rates and state benefit maxima that the Department of Labor compiles in its comparison of state UI laws (`https://oui.doleta.gov/unemploy/statelaws.asp`).

κ_V **and** κ_r**, the voucher rules.** These two set the income cutoff for eligibility and the tenant's rent contribution, respectively. Raise κ_V and more households are eligible, so more have a voucher to lose under Treatment First; raise κ_r and a voucher pays less, so there is less to lose from losing a voucher. I take both from HUD's own rules. I set them to half of median income and thirty percent, respectively. The first is the very low-income limit HUD publishes annually with its income limits (`https://www.huduser.gov/portal/datasets/il.html`); the second is the share of its before-tax income a tenant pays towards the rent, the voucher paying the rest up to the payment standard, under the Housing Choice Voucher programme (`https://www.hud.gov/helping-americans/housing-choice-vouchers`).

ϱ**, the food stamp clawback rate.** This is the rate at which food assistance is withdrawn as income rises. Raise it and food assistance is clawed back faster, so it stops at a lower income and pays less to every household that still receives it. I set it to thirty percent, which is the statutory rule (7 CFR 273.10).

Block 13: Drug market. The two parameters in this block describe the rest of the drug market, closing the loop on the elasticity explained in Block 6: the first, J_E, explained in footnote 15, fixes who can enter it, and the second is set so that supply equals demand in the benchmark, and is then held fixed.

ν_1**, the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark, and kept fixed across experiments, in which the price is free. It normalises units, not behaviour.

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